Bt102

Microbiology

FINAL TERM OBJECTIVE + SUBJECTIVE

Admire Group of Zoologist

GROUP PROPERTIES

- Separate Group For Each Subject
- Separate Group For Discussion
- vedio Lectures
- Ideal Assignments
- Ideal gdbs

اس گروپ میں ویڈیو لیکچرز، آئیڈیل اسائنمنٹس
اور آئیڈیل GDB مہیا کی جاتی ہے

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Also Available

Designed by Shakir

1st Excellent Zoologist Group for 1st Semester
And Lion Group for 2nd Semester

BT-102-Microbiology

From 53 to104 lessons

Final term all types questions

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PREARED BY: ABRAR HUSSAIN

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1-Bacterial Identification

01-Proteins from microbes are separated by electrophoresis.
02-Protein bands are blotted onto a nitrocellulose membrane.
03-Proteins could be detected by electrophoresis and blotting.
04-Blots can be detected by specific antibodies labeled with radioisotopes etc.
05-Nucleic acid base microbial detection methods are three.
06-Nucleic acid base microbial detection methods are DNA probes, DNA chip and PCR.
07-DNA probes can be used for diagnosing various diseases.
08-DNA chip can be manufactured to contain hundreds and thousands of synthetic single-stranded DNA sequences. Assume that each DNA sequence was unique to a different gene.
09- After 30 cycles, enough targets DNA accumulate which could be detected by electrophoresis.
10- PCR provides a quicker way of identification of microbes.

2-Fungi

11-Fungi are chemoheterotrophs.
12-Fungi acquire nutrients by absorption.
13-Two types of reproductions asexual and sexual reproduction are present in fungi.
14-Hyphae is thread like filaments of a fungus
15-Mycelium is collection of hyphae.
16-Thallus is collection of mycelia.
17-Mold is filamentous fungi.
18-Yeast is unicellular fungus.
19-Mycology is the branch of biology which deals with the study of fungi.
20-There are two types of hyphae of fungus septate and non-setate or coenocytic.
21-The septate hypae are have cross-walls and divide into uninucleate cell-like units.
22- Vegetative hypha are absorbs nutrients.
23- Aerial or reproductive hypha are bears reproductive spores.
24- Fungi are eukaryotes and multicellular except yeasts.
25-There are three types of fungi spores' asexual spores, conidiospores and sporangiospores.
26 - Asexual spores are formed by mitosis of hyphal cells.
27- Conidiospores are unicellular or multicellular spore not enclosed by a sac.
28-Sporangiospores are enclosed by a sac.
29-There are four types of conidiospores, conidia, arthroconidia, blastoconidia and chlamydioconidia.
30- Conidia are chain of spores at the end of a conidiphore Aspergillus.
31- Arthroconidia are spores formed by fragmentation of septate hypha of a coccidioides immitus.
32- Blastoconidia are spores as budding of the parent cells of a candida albicans.
33- Chlamydioconidia are spores formed by rounding and enlargement within a hyphal segment of a candida albicans.
34- Sporangiospores are formed within a sporangium or sac at the end of an aerial hypha called sporangiophore of rhizopus.
35- Plasmogamy is a haploid nucleus of a donor cell (+) penetrates the cytoplasm of a recipient cell (-).
36-Karyogamy is the (+) and (-) nuclei fuse to form a diploid zygote nucleus.
37-Meiosis is the diploid nucleus gives rise to haploid nuclei (sexual spores).
38-Yeasts are non-filamentous fungi, unicellular and reproduce by budding (asexually) or sexually but some are dimorphic.
39-Yeast has 37oC optimum temperature for growing.
40–Molds have room temperature for growing.
41-Yeasts are larger or bigger than bacteria, no flagella or cilia.
42-Yests are facultative anaerobes are larger than bacteria and are gram positive.
43- Aspergillus and Trichophyton are filamentous fungi.
44-Yeast can ferment and can produce various products.
45-Biochemical testing is used for yeast identification.
46-Morphology is used to identify filamentous fungi.
47-Saccharomyces are used for alcoholic beverages.
48-Anaerobic conditions are used for alcohol.
49-Aerobic conditions are used for bread making.
50-Yeast used in biotechnology for making hepatitis B vaccine.
51-Aspergillus niger is used for making citric acid.
52-Trichoderma (plant cell digestion to clear fruit juices) used for making cellulase enzyme.
53-Taxomyces (anticancer) used for making anticancer drug texol.
54-Entomophaga (kills gypsy moth) is used for biological pest control.
55-Penicillium is used for making antibiotic penicillin.
56-Mycosis is a general term used for fungal diseases.
57-Systemic mycosis is found deep within the body and histoplasmosis and coccidioidomycosis are its examples.

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58-Subcutaneous mycosis is found beneath the skin.
59-Sporotrichosis is found in gardeners and farmers and it is example of subcutaneous mycosis.
60-Trichophytons use keratinase to degrade keratin in nails, hair and skin and include ringworm infections are examples of Cutaneous Mycosis.

3-Lichen

61-Combination of green alga or a cyanobacterium (phycobiont) and a fungus (Mycobiont) is called Lichen.
62- Lichens have symbiotic relationship.
63- Lichens partners do not grow alone.
64- Lichens grow where no other microbes can grow.
65- Lichens grow on bare rocks, tree trunks and house roofs and bare soils.
66-Typically lichens grow extremely slowly as 1 mm to 3 cm per year.
67-Lichens are classified on the basis of the fungus.
68-There are three types of lichens, crustose, foliose and fruticose.
67-Crustose are forms a thin crust.
68-Foliose are leaf like.
69-Fruticose are finger-like projections.
70-Lichens thallus is devide into three parts the medulla, cortex and rhizine.
71-Medulla is algal layer and fungal hyphae.
72-Cortex is could be two upper layer and lower layer.
73-Rhizine is roots or holdfasts.
74 Lichens are used to make dyes for clothes.
75-Lichens are used to make usnic acid as antiseptic in China.
76-Lichens cause allergic contact dermatitis.

4-Algae

77-Algae are simple eukaryotic cells, unicellular, some others are multicellular (thallus).
78-Algae have lack tissues (roots, stem and leaves) seen in plants absorb nutrients from water through their surfaces.
79-Algae are mostly photoautotrophs, a few are chemoheterotrophs.
80- Algae are mostly aquatic (water is necessary for physical support, reproduction and diffusion of nutrients).
81- Alage cell covering thallus can carry photosynthesis.
82- Some algae float by gas-filled bladders.
83- Their location in oceans depends on the availability of nutrients, wavelength of light and the surface on which to grow.
84- The body of multicellular alga is called a thallus which consists of branched holdfasts (anchor alga to rock) stem like hollow stipes and leaf like blades.
85- All algae can reproduce asexually.
86-The multicellular algae fragment can form a new thallus and mitosis followed by cytokinesis in unicellular algae.
87-The alternation of generations of haploid and diploid individuals is found in algae during sexual reproduction.
88-The algae Phaeophyta (Kelp) are brownish in color with cellulose and alginic acid cell walls.
89-The algae Phaeophyta (Kelp) are multicellular with chlorophyll a and c, xanthophylls.
90-The algae Phaeophyta (Kelp) store carbohydrates and harvested for align.
91-The algae Rhodophyta are reddish in color with cellulose cell walls.
92-The algae Rhodophyta are multicellular with chlorophyll a and d, phycobiliproteins.
93-The algae Rhodophyta store glucose polymerand and harvested for agar and carrageenan.
94-The algae Chlorophyta are green algae with cellulose cell walls.
95-The algae Chlorophyta are unicellular or multicellular with chlorophyll a and b.
96-The algae Chlorophyta store glucose polymer and rise to terrestrial plants.
97-The algae Bacillariophyta are brownish in color with pectin and silica cell walls.
98-The algae Bacillariophyta are unicellular with chlorophyll a and c, carotene, xanthophylls.
99-The algae Bacillariophyta are Store oil.
100-The algae Bacillariophyta are fossilized diatoms formed oil and also produce domoic acid.
101-The algae Dinoflagellata are brownish in color with cellulose in plasma membrane.
102-The algae Dinoflagellata are unicellular with chlorophyll a and c, carotene, xanthins.
103-The algae Dinoflagellata store starch.
104-The some Dinoflagellata are symbionts in marine animals.
105-The Dinoflagellata are produce neurotoxins cause paralytic shellfish poisoning.
106-The algae Oomycota are colorless or white with cellulose cell walls.
107-The algae Oomycota are multicellular and chemoheterotrophic.
108-The algae Oomycota produce zoospores.
109-The algae Oomycota are decomposers and plant parasites.
110-The Phytophthora infestans are responsible for Irish potato blight.
111-The P. cinnamomi infects eucalyptus and P. ramorum causes sudden oak death and both are decomposers and plant parasites.
112-The algae are important part of aquatic food chain.
113-The algae fix CO2 into organic compound.
114- 80% earth O2 produced by planktonic algae.
115-Petroleum and natural gas are the residues (hydrocarbons) of diatoms and other planktonic organisms.
116-The unicellular algae are symbiont in animals.

5-Virus

117-Infectious viral particle, composed of nucleic acid, surrounded by a protein coat called capsid.
118- Viruses have an extracellular and an intracellular state.
119-The outside of a cell, in the extracellular state, a virus is called a virion.
120-Viroid are plants pathogens and infectious circular RNA of plants.
121-Viroid are formed of only RNA.
122-The protein coat is absent in viriod.
123-Viroide is smaller in size than a typical virus and infects plants only.
124-In virus structure the nucleic acid either DNA or RNA (never both) Single or double stranded, circular or linear with one segment or multiple segments.
125-The capsid of virus is made up of protein subunits can be single type or multiple types called capsomeres.
126-The Arrangement of capsomeres is characteristic of a particular type of virus.

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127-In some viruses, the capsid is covered by an envelope (lipid, protein and carbs).
128-In virus may or may not have spikes (carbohydrate-protein complexes) projecting from the surface of the envelope (lipid, protein and carbs).
129-Most viruses infect only specific types of cells in one host.
130-Host range of virus is determined by specific host attachment sites and cellular factors.
131-Viruses are helical, resemble long rods, rigid or flexible.
132-Rabies virus is icosahedral and takes polyhedral shapes.
133-Icosahedron mean regular polyhedron with 20 triangular faces and 12 corners and the poliovirus is its example.
134-The Influenza virus is enveloped virus and roughly spherical.
135-Complex viruses head is polyhedral with N.Acid , tail sheath is helical and the bacteriophage is its example.
136-In viral taxonomy the family names end in viridae and genus names end in virus.
137-Viral species are group of viruses sharing the same genetic information and ecological niche (host).
138-Herpesviridae is family of herpesvirus.
139-In viruses the common names are used for species and subspecies are designated by a number.
140-HHV-1, HHV-2 and HHV-3 are human herpes virus.
141-Viruses attach to cell membrane.
142-Host cell receptors (determines the range of infectivity).
143-Penetration by pinocytosis (plasma membrane folds inwards to from a vesicle) is called an entry.
144-Viral envelope fuses with plasma membrane and releases the capsids into the cell's cytoplasmic fusion.
145-Separation of viral DNA from its protein coat by viral or host enzymes is uncoating (undergoes eclipse period).
146-Production of nucleic acid and proteins is called biosynthesis.
147-Nucleic acid and capsid proteins assemble is maturation.
148-Viruses release by budding (enveloped viruses) or rupture.
149-Biosynthesis is early proteins, late proteins, genome replication, assembly and release.
150-Lysis of cells, microscopic and macroscopic degenerative changes (Cytopathic Effects) are consequences of viral infection.
151-Virus does not replicate is called latent infection.
152- Virus replicates but slowly is called persistent infection.
153-The viral transformation is transference of tumor cells.
154-Cultivation of viruses include need live cells to grow, animals, embryonated eggs, cell cultures and plants.
155-Plaques formation is equivalent to bacterial colony.
156-Plaque is an area where the virus infection has lysed the cells.
157-In virus enumeration plaque assay is plaque forming units are counted, infectious dose 50% (ID50) – TCID50 and lethal dose 50% (LD50).
158- Virusoids are also called satellite RNA with circular ssRNA .
159-Virusoids are encode one or two genes products and need helper virus to infect host cells. The Human hepatitis D virusoid uses hepatitis B virus as helper.
160- Prions are proteinaceous infectious particles, abnormal form of cellular proteins and cause neurodegenerative diseases such as scrapie in sheep, bovine spongiform encephlopthy and kuru in humans.
161-Genetic component involved in prions.

6-Epidemiology

162-Epidemiology is the study of where and when diseases occur and how they are transmitted in a population.
163-There are two epidemiology methods the descriptive (recording the data about disease) and analytical (To determine the cause of disease).
164-Descriptive method includes location and time of cases of disease, gender, health, age of patients etc.
165-Analalytical method includes analysis of data, mode of transmission and means to prevent disease and the experimental epidemiology, hypothesis of a disease is tested and application of Koch's postulates.
166- Epidemiology case reporting includes health care workers report specified disease to local, state, and national offices, incidence and prevalence of disease, origin and spread of a disease and data helpful for control measures.
167-Morbidity is incidence of a disease.
168-Morbidity rate is the number of people affected in relation to the total population in a given time period.
169-Mortality is deaths from a disease.
170-Mortality rate is the number of deaths from of a disease in relation to the population in a given time period.
171-Pathogenicity is the ability to cause disease.
172-Virulence is the extent of pathogenicity.
173-The learning Objectives of epidemiology are language of disease, microbiota, factors that determine the microbiota , symbiosis, microbial antagonism, description of diseases, stages of diseases and sources of infection.
174-Pathology is the study and causes of diseases.
175-Pathogenesis is how disease establishes itself and structural and functional changes.
176-Infection is invasion by the microbes.
179-Disease is a dysfunction.
180-Microbiome is microbial communities that live in or on humans.
181-Normal microbiota or flora is permanent residence but do not cause diseases under normal conditions.
182-Transient microbiota is present for several days, weeks or months and then disappears.
183-Physical and chemical factors of microbiota are pH, O_2 and CO_2.
184-Microbiota also includes nutrients and its host defense includes stress and age.
189-Symbiosis means living together.
190-In mutualism both organisms are benefited.
191-In parasitism one organism is harmed.
192-In commensalism one organism is benefited, other is unharmed.
193-Microbial antagonism is a competitive exclusion, compete for nutrients, by chemicals, acteriocins by E. coli suppresses Salmonellae.
194-Clostridium difficile is normal microbiota keeps it low but antibiotics can result in diarrhea.
195-Opportunistic pathogen is normal microbiota under stress conditions can cause infections.
196-Endemic is consistently present, but with low incidence and infected ones are reservoirs of infection.
197-Epidemic within an area is unusually high incidence of a disease.
198-Pandemicis is worldwide epidemic.
199-Incidenceis number of new cases in a given time period of a population.
200-Prevalence is new + existing cases.
201-Infection is colonize grows in body.
202-Incubation period is time b/w infection and diseases.
203-Acute period is height of disease.
204-Decline period is disease symptoms subside.

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205-Convalescent period is patient returns to normal.
206-Reservoir is sites where organisms of infections remain viable and act as source of infection.
208-Animate is live in the body spread from person to person by direct or indirect contact.
209-Single host organisms can be eradicated or controlled.
210-Zoonosis is primarily an animal infection, but can be transmitted to humans, assume epidemic proportions in humans and difficult to control or eradicate.
211-A living carrier is a pathogen infected individual who does not show any signs or symptoms of the disease, potential source of infection for others and typhoid Mary story.
212-Direct contact transmission is person to person and a physical contact i.e Influenza.
213-In indirect contact a nonliving object is involved.
214-Fomite is any nonliving object (towel, bedding).
215-Droplet is mucus droplets.
216-Vehicle Transmission is transmission by a medium, waterborne, foodborne, airborne, blood and other body secretion.
217-Vectors are arthropods.
218-Mechanical transmission is (Passive) and biological transmission is (Active).

7-Pathogenicity

219-Portals of entry are mucus membrane, tubal organs, respiratory system and digestive system as largest portals.
220-Skin is the largest organ of the body in terms of the surface area.
221-Parenteral route is by injury and preferred route of entry is by streptococci inhaled vs ingested.
222-Invading Organisms are few vs many in number.
223-Virulence can be expressed in terms of ID50 and depends upon the route of entry.
224-Bacillus Anthracis accourding to skin ID50 is 10-50 spores, inhalation is 10 -20 thousands spores, GI anthrax ID50 is 250 to 1000 thousands spores.
225-Potency of toxin is expressed in LD50.
226-Botulinum toxin in mice is 0.03 ng/kg.
227-Shiga toxin is 250 ng/kg.
228-Staph enterotoxin is 1350 ng/kg.
229-Adhesion is mediated by adhesins or ligands to the receptors and present on glycocalyx, pili, fimbriae, or flagella also these are glycoproteins or lipoproteins.
230-The dental plaques is an example of biofilm.
231-The capsule impairs phagocytosis and antibodies can help in defense evasion.
232-Strep pneumoniae and Klbesiella pneumonia are few bacteria produce capsule.
233-Mycolic Acid and M protein found on cell surface and fimbriae and resists phagocytosis are components of pathogen cell wall.
234-(Exoenzymes) coagulase can blood clotted and bacteria is trapped or hides itself in the clot also staphs produce it.
235- Hyaluronidase enzyme dissolves CT and bacteria can spread easily.
236-Collagenase enzyme breaks down collagen.
237-IgA protease enzyme digests IgA antibodies.
238-Influenza virus and Neisseria gonorrheae are antigenic variations.
239-Nutrient depletion and siderophores for iron is damage by microbes and are lactoferrin, transferrin ,ferritin.
240-Bacteria that grow in host cells rupture the cells and spread further is called direct damage.
241-Toxin produces exotoxins, antitoxins, toxoid, A-B toxins.
242-Superantigens are responsible of provoke intense immune response and cytokines are released by T cells.
243-There are two types of toxins, A-B toxins and Membrane disrupting toxins.
244-The diphtheria toxin is type of A-B toxins.
245-Membrane disrupting toxins cause lysis of cells, Leukocidins and Hemolysins.
246-Endotoxins gram –ive bacteria cause fever, shock, limulus amebocyte lysate assay (LAL) and hemolymph from horseshoe crab.

8-Innate Immunity

247-Immunity is the ability of the body to protect itself from harmful effects of diseases.
248-Susceptibility is lack of immunity.
249-There are two types of immunity the Innate and Adaptive.
250-Innate immunity is present at birth.
251-Adaptive immunity acquired by experience.
252-Innate Immunity is present at birth, rapid to respond, nonspecific, do not increase in response with repeated exposures to the same agent, have no memory and have first line of defense: Physical barriers (skin, mucus membrane),
Second line of defense: Phagocytic cells and their secretions.
253- Adaptive Immunity is a specific, have memory, increase with repeated exposure, slower to respond, humoral immune response, cellular immune response.
254-Toll like receptors (TLRs) are protein molecules on defensive cells.
255-Pathogen associated molecular patterns (PAMPs) are formed by LPS, flagellin, peptidoglycan, DNA, RNA.
256-Cytokines are released as a result of defense cell stimulation by PAMPs and TLRs interactions.
257-Cytokines recruits more immune cells and also stimulate adaptive immune response.
258-The physical factors of 1st line of defense are skin, mucus membrane (mucus), lacrimal apparatus and cilia.
259-The other physical factors are urine, earwax, peristalsis, vomiting, diarrhea and defecation.
260-The chemical factors of 1st line of defence are cebum, pH of the skin (3-5), lysozyme in sweat, tears, saliva, body fluids and urine and acidic pH of stomach.
261-The 2nd line of defense contains leukocytes and lymphocytes cells.
262-Leukocytes contain granulocytes and agranulocytes cells.
263-Lymphocytes contain natural killer, B cell and T cell.
264-Granulocytes contain neutrophils, eosinophils and basophils cells.
265-Agranulocytes contain monocytes cells.
266-Inflammation is damage to body tissues triggers a local defensive response,the redness, pain, swelling and heat are its symptoms.
267-Cytokines are released (TNF-alpha) that causes release of acute phase proteins from liver.
268-Fever is systemic response, controlled by hypothalmus, pyrogens by microbes, IL-1, TNF alpha and growth of microbes arrested.
269-More than 30 difference proteins make complement system.
270-In complements the cells of the immune system are destroying microbes.
271-Interferons are antiviral proteins produced by macrophages and lymphocytes in response to viral infections (fibroblasts) and interfere with viral replication, species specific and not viral specific.
272-Types of interferons are alpha (infected cells), beta (infected) and gamma (lymphocytes).
273-Antimicrobial Peptides are 12 - 50 amino acid long peptides.
274- Antimicrobial Peptides are defensins by neutrophils, dermicidin by sweat glands and thrombocidins by platelets.
275- Antimicrobial Peptides are secretions of innate immunity, cytokines, complement and interferones.

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9-Adoptive Immunity

276-Adoptive immunity is an immunity that occurs after exposer to an antigen either from pathogen or a vaccination.
277-There are three lines of defense of adoptive immunity.
278-1st line of defense is skin.
279-2nd line of defense is macrophages, neutrophils.
280-3rd line of defense is B and T cells.
281-B cell are responsible for humoral immunity.
282-T cells are responsible for cellular immunity.
283-Antigen is a foreign substance that provokes immunity.
284-More specifically antigen called as Immunogen is a substance that produces immune response.
285-Epitopes is a specific regions on an antigen and also called as antigenic determinants.
286-Most antigens have MW >10,000 and the < 10,000 D molecules are poor antigens – Haptens is example.
287-Antigen needs a carrier protein to make antibodies.
288-Once Abs is formed when they can bind hapten– Penicillin is example.
289-Antibodies are glycoproteins in nature and present in gamma globulins of serum.
290-The antibodies IgM in blood, IgG in serum and body secretions, IgA of (mucus membrane) and IgE is protect from (allergy) and IgD in plasma membrane.
291-There are five classes of antibodies.
292-Each B cell has unique antibody molecules against one epitope only and B cell never changes its specificity.
293-B cells in bone marrow create antibodies without even seeing the antigens and it is like choosing a shoe or a shirt from a shop.
294-Ag-Ab binding respond agglutination, opsonization, complement activation, Ab-dependent cell-mediated cytotoxicity and neutralization.
295-In MHC (Major Histocompatibility complex) the MHC I is Present on all nucleated cells, MHC II is Present only on APCs and Antigen presenting Cells.
296-T helper cells are TH1 (CD4+) and TH2 (CD4+).
297-T cytotoxic cells are (CD8+).
298-APCs are dendritic cells, macrophages and B cells.

10-Special staining

299-In Spore stainings the spores contain dipicolinic acid + Ca++ and spores become heat resistant.
300-So, strong stains are used with heat and make a smear and air dry it.
301-In Spore stainings the pour malachite are green on the slide.
302-Spore stainings keep it there on the boiling water beaker for 5 min and wash the smear with dH2O.
303-The observetion of spore staining under and by oil-immersion lens.
304-In flagella stain, transfer a drop of fresh culture on one end of a slide, allow the drop to trickle down slowly, air dry the smear and do not heat fix it. Cover the smear with mordant (tannic acid 10% in 5% NaCl) for 10 to 30 min, rinse the smear gently with H2O, add carbol fuchsin and let it sit for 5-15 min, rinse off the stain with water , air dry the smear and see under the microscope.
305-The flagella can be seen with or without the stain (by observing motility).
306-Acid Fast Staining is a differential stain .
307-In 1882, Paul Ehrlich observation on Mycobacterium.
308-In acid fast stained do not loose the stain even with acid alcohol.
309-Mycobacterium have mycolic acid (waxy).
310-The Ziehl Neelsen method is commonly used in acid fast staining.
311-In acid fast staining the stain wash with water and then with acid alcohol for 1 min and add counterstain with methylene blue for 1 min.
312-Acid fast bacteria have mycolic acid that can be stained by heating the organisms with the stain.
313-The capsular staining is extracellular material (glycocalyx).
314-The regular and thick structure is called capsule.
315-The loose and not organized structure is called slime.
316-The thickness of the capsule depends upon the culture conditions.
317-The capsular organisms use and make the broth viscous and stringy.
318-The capsular organisms colonies look moist, glistening, and sticky and involves in virulency.
319-In capsule stain prepare a thick smear in a loopful of congo red (1%) stain, fixation in acid alcohol for 15 sec ,Wash with dH2O, cover it with acid fuchsin for 1 min and again wash with water .
320-In capsule stain the bacteria stain is red, colorless capsule with dark blue background.

11-Hypersensitivity

321-The abnormal antigenic response is called hypersensitivity and its other name is allergy.
322-Those antigens that are causes hypersensitivity or allergy called allergens.
323-There are four types of hypers, type I (Anaphylactic), type II (Cytotoxic), type III (Immune complex) and type IV (delayed cell-mediated).
324-Anaphylaxis is difficulty in respiration, Shock, hypotension and increase in heart rate.
325-The hyper anaphylactic happens in 30 min, IgE is involved which binds to mast cells and the mast cells degranulate and secretes mediators of inflammation.
326- Anaphylaxis is shock from drug reactions venoms, common allergens causing asthma, diagnosis by demonstration of IgE by i/d injections of allergens one by one, quantitation of IgE by ELISA.
327-Rx is epinephrine & antihistaminic drugs.
328-Type II cytotoxic involves in antigenic cell, IgG or IgM and complement.
329-Blood transfusion reaction is an example type II cytotoxic.
330-In drug induced cytotoxicity the cell deficiencies, low number of RBCs due to hemolysis, thrombocytopenia if platelets are diagnosis.
331-IgG or IgM involved in causes damage to the cells and treated according to the cause.
332-Immunne complex is antigen-antibody complex involves in soluble antigen circulating in the serum, IgG or sometimes IgM and Complement.
333-Immune complexes are formed which when get stuck in capillaries cause damage by compliment activation.
334-The type IV (delayed cell-mediated) is originally called tuberculin reaction and it is involved in antigen, T cells and macrophages.
335-T helper cells are involved mostly to avoid the contact or symptomatic Rx.

12-Vaccine

336-Vaccine is suspension of organism or fraction of organism that induces immunity.
337- Vaccine mimics the infection by the organism but without causing the disease is its principle.
338-There are several types of vaccines including inactivated vaccines, live attenuated vaccines, mRNA vaccines, subunit, recombinant, polysaccharide and conjugate vaccines, toxoid vaccines and viral vector vaccines.
339-live attenuated vaccines use a weakened from of the germ that causes a disease.
340-The better immunity , long term immunity and Virus replicates increasing the Ag dose are benefits of live vaccines.
341-Inactivated vaccines killed organism with formalin or phenol.

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342-Inactivated vaccines killed rabies, polio, Influenza and hepatitis A germs and are mostly induce humoral immunity.
343-Toxoids are inactivated toxins and used as a vaccine.
344-Conjugated vaccines are poor antigens attached to proteins.
345-DNA vaccines are injected into cells and these proteins are expressed which stimulate immunity.
346-Polysaccharide vaccines have poor success.
347-Subunit vaccines are portion of viral or bacterial proteins that can induce immunity and also called recombinant vaccines.
348-Organisms in bulk are needed to make vaccines.
349-Bacteria can be grown in flasks by providing medium of growth.
350-Cell culture is used for viral vaccines.
351-Chicken embryos can be used to grow viruses such as Influenza vaccine.
352-Animals can be used to grow viruses.
353-Plants can be used in recombinant technology.
354-Vaccines mimic infection but without harmful effects and various types are available.
355-Booster is required for better protection.

13-Antimicrobials

356-Paul Ehrlich coined the word "chemotherapy". It means the use of chemical agents to treat diseases.
357-Paul Ehrlich observation is the organisms stained without staining the background.
358-In 1928 Alexander Fleming working with Staph Aureus observed colony of a mold with inhibitory effect and invent penicillin from Penicillium Notatum.
359-The antimicrobials are used for chemotherapy and selectively toxic to microbes sparing body cells.
360-Many antibiotics are produced by microbes, but only a few are selectively toxic to microbes.
361-Antibiotics are products of microbes against microbes and must not be toxic to the body.
362-Developing drugs against prokaryotes are relatively easy and against eukaryotes are difficult.
363-Narrow targets only a group of organisms – Penicillin for G+ve and Narrow well targeted – Spare microflora.
364-Wide targets many groups of microbes – Tetracycline. Targets every organism – Destroy normal microflora as well.
365-Penicillins, Cephalosporins, Bacitracin and Vancomycin target the cell wall.
366-Chloramphenicol , Erythromycin , Tetracyclines and Streptomycin target protein synthesis.
367-Quinolones and Rifampin target DNA replication.
368-Polymixin B target plasma membrane.
369-Sulfanilamide and Trimethoprim target metabolites.
370-Antibiotics are selectively inhibit various components of microbes.
371-The bactericidal antibiotics are kill microbes directly.
372-The bacteriostatic antibiotics prevent growth of microbes.
373-When target is peptidoglycan the cell wall does not get synthesized, cell lysed as a result (cidal) and targets growing cells.
374-The no protein synthesis is based on ribosome structural differences in prokaryotes and eukaryotes.
375-Eukaryotes ribosome is 80S and Prokaryotes ribosome is 70S.
376-Chloramphenicol binds to 50S and Inhibits peptide bonding and is bacteriostatic.
377-Streptomycin changes shape of 30S, misreading of RNA codes and is bacteriostatic.
378-Tetracyclines are interfere with attachment of tRNA to mRNA ribosome complex and is bacteriostatic.
379-Polypeptide antibiotic brings changes in the permeability of plasma membrane and result in the loss of imp metabolite from the cell.
380-Amphotericin B and Ketoconazole are examples of antifungal drugs.
381-Bacteria lack sterol.
382-Plasma membrane combines with sterols.
383-Sulfanilamide and para-aminobenzoic acid (PABA) are essential metabolites.
384-Nucleic Acid Synthesized during DNA replication and transcription.
385-The very limited b/c can interfere with body cells.
386-PABA is a substrate for an enzyme that is involved in folic acid synthesis.
387-FA is needed for purine and pyrimidine synthesis.
388-Physicians use prescribes drugs that they know will work.
389-If the organism has been identified, selection of drug is easy.
390-In Disk Diffusion Method the standardized inoculum spread on agar plate, filter paper disks soaked with antibacterials, zone of inhibition is recorded and zone compared to a standard table.
391-The drugs are sensitive, intermediate and resistant according to report.
392-The abbreviation of (MIC) is Minimum Inhibitory Concentration.
393-The abbreviation of (MBC) is Minimal Bactericidal Concentration.
394-The lowest concentration that inhibits visible growth.
395-The various dilutions of the drug made and tested with test organism.
396-If the growth occurs; the drug is bacteriostatic and otherwise bactericidal.
397-The susceptibility of organisms is recorded over a period to see if there is a change in response.
398-Cell wall breakage antibiotics are penicillin, cephalosporins, polypeptide antibiotics anti antimycobacterials.
399-The Cefixime is (4rth generation) Cephalosporinantibiotic.
400-Polypeptide antibiotics are bacitracin (topical) and ancomycin (Pen-R).
401-Chloramphenicol (potentially cytotoxic), Aminoglycosides, Tetracyclines, Tetracyclines and Macrolides are protein synthesis antibiotics.
402-Erythromycin and Azithromycin (ssyn) are Macrolides.
403-Plasma membrane breakage antibiotic is Polymixin B and use for topical G-ve bacteria.
404-Tetracyclines are used for gainst chlamydias and rickettsias.
405-Rifampin, Quinolones and Flouroquinolones are nucleic acid synthesis antibiotics.
406-Rx (Rifampin) is for tuberculosis.
407-Nalidixic acid, Norfloxacin and Ciprofloxacin are examples of Quinolones and Flouroquinolones and used preferably for urinary tract infections.
408-Drugs can selectively kill fungi and viruses butt not all viruses could be killed by drugs.
409-The various antibiotics target various components of the cell.
410-Polyenes and Azoles are Fungal Sterols (PM).
411-The Polyenes; Amphotericin B is used for systemic infections of fungi.
412-Clotrimazole (topical), Ketoconazole, Voriconazole are Azoles.
413-Crosses BBB is Voriconazole.
414-Flucytocine inhibits synthesis of RNA and DNA in fungal cells and it is nucleic acid synthesis drug.
415-Griseofulvin is an antifungal drug that inhibition of mitotic microtubules.
416-Nucleoside and Nucleotide analogs are antivirus drugs.
417-Acyclovir, Ganciclovir, Ribavirin are nucleoside and nucleotide analogs used mostly for herpesviruses.
418-Attachment and uncoating drugs of (flu virus) are Amantadine, Zimantadine and Zanamivir that inhibits Neuraminidase.

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Thanks to all excellent groups admins

419-Chloroquine and Metronidazole are antiprotozoan drugs.
420-Chloroquine inhibits DNA synthesis and used against malaria.
421-Metronidazole interferes with anaerobic metabolism and used for Giardiasis, Amebiasis and Trichomoniasis.
422-Praziquantel, Albendazole, Mebendazole and Ivermectin are anti-helminthic drugs.
423-Praziquantel alters permeability of PM and used for tapeworm and fluke infections.
424-Albendazole, Mebendazole inhibit absorption of nutrients and used against intestinal roundworms.
425-Ivermectin paralyzes worms and used against intestinal roundworms, mites and lice.
426-Protozoa and helminths can be killed too with drugs.
427-Drugs resistance begins with mutations and transmitted by conjugation and transduction.
428- Blocking Entry, inactivating by enzymes, alteration of target molecule and efflux of antibiotics are further drugs resistance.

FINAL TERM SOLVED QUESTIONS

1. Karyogamy plasmogamy and meiosis? 2MARKS (PPT 58)

Plasmogamy: A haploid nucleus of a donor cell (+) penetrates the cytoplasm of a recipient cell (-)
Both cells are haploid (one set of chromosomes).
Karyogamy: The (+) and (-) nuclei fuse to form a diploid zygote nucleus.
Meiosis: The diploid nucleus gives rise to haploid nuclei (sexual spores).

2. Advantage of Dry Heat relatively? 2 MARKS

Advantages of dry heat sterilization include: it is inexpensive, it does not pollute or cause toxic fumes, and it does not corrode or rust metal objects.
Disadvantages include: it is relatively slow and many objects cannot withstand the high temperatures

3. How Microorganism cause damaged the body?

Infection with a **pathogen** does not necessarily **lead** to disease. Infection occurs when viruses, **bacteria**, or other **microbes** enter your **body** and begin to multiply. ... Pathogenic **microbes** challenge the immune system in many ways. Viruses make us sick by killing cells or disrupting cell function.

4. Microbial Identification?

Microbial identification can be **defined** as "**microbial** characterization by a limited spectrum of tests pre-chosen and appropriate to the problem being studied"
Methods: Classical or Conventional Method, Serological testing, Nucleic acid based testing.

Immune and Its Types? Define Imunity and Its types? Innate and adaptive immunity? (10)

Immunity is the balanced state of multicellular organisms having adequate biological defenses to fight infection, disease, or other unwanted biological invasion, while having adequate tolerance to avoid allergy, and autoimmune diseases. Innate Immunity
Plants and animals have what is called innate immunity. Innate immunity is the first line of defense against pathogens. It involves several cell types, proteinand even an organ. The organ involved is your skin. Yes, skin is part of the firs line of defense. It protects you and prevents pathogens from getting inside your body.
Adaptive immunity is an important part of the immune system. It is protection from an infectious disease agent that is mediated by B- and T-lymphocytes following exposure to specific antigen, and characterized by immunologica memory.

7. Economies use of fungi?

They play an important role in medicine yielding antibiotics, in agriculture by maintaining the fertility of the soil and causing crop and fruit diseases, forming basis of many industries and as important means of food.

8. Gram Staining Procedure?

Crystal violet is added. – Primary stain • Crystal violet is washed off. • Iodine is added to enhance binding. – Mordant. • The slide is washed off with alcohol. – Decolorizing agent • Gram positive bacteria retain crystal violet • Gram negative b acteria appear colorless. • Crystal violet and iodine make a complex in the cytoplasm • Peptidoglycan layer is thicker in Gram positive bacteria and CV-I is retained. • Alcohol is rinsed off and stained with safranin. – Counterstain • The smear is washed again. • Blotted dry and examined microscopically.

9. Difference Simple and Differential Media?

Selective and differential media are used to isolate or identify particular organisms. Selective media allow certain types of organisms to grow, and inhibit the growth of other organisms. ... Differential media are used to differentiate closely related organisms or groups of organisms

10. Horizontal and Vertical Gene Transfer?

Vertical gene transfer – Parents to offsprings
Horizontal – From cells to cells within the same generation – From bacteria to bacteria.

11. Difference LB Broth and LB Agar?

LB Broth: **broth** (**LB**) is a nutritionally rich **medium** primarily used for the growth of bacteria. Its creator, Giuseppe Bertani, intended **LB** to stand for lysogeny **broth**, but **LB** has also come to be commonly referred to as **Luria broth**,
Lennox **broth**, or**Luria**-Bertani **medium**.
LB Agar:

Luria broth (**LB**) is a nutrient-rich media commonly used to culture bacteria in the lab. The addition of **agar** to **LB** results in the formation of a gel that bacteria can grow on, as they are unable to digest the **agar** but can gather nutrition from the **LB**within.

12. Chemical used in Gram staining?

Both gram-positive and gram-negative cells have peptidoglycan in their cell walls, so initially, all bacteria stain violet. Gram's iodine (iodine and potassium iodide) is applied as a mordant or fixative. Gram-positive cells form a crystal violet-iodine complex. Alcohol or acetone is used to decolorize the cells.

13.Viruses, Virion, and Virusoid and Prion?

Viruses: Acellular organisms – Proteins and nucleic acids
Viroids: composed only of circular ssRNA (potato spindle tuber viroid)
Virusoids: Contain circular ssRNA, need helper viruses for replication and encapsidation; also called satellite viruses
Prions: Infectious proteins

14.Uses of UV in Lab?

• Transilluminators. **Used** for the visualization of molecular samples, transilluminators are indispensable tools for researchers at life-sciences **laboratories**. ...
• **UV** Crosslinkers. ...

• Fluorescence Analysis Viewing Cabinets and Work Stations. ...
• **Ultraviolet** Lamps. ...
• Digital Radiometers/Photometers

15.Epidemiology and Its type?

Monitoring diseases in a population • Detection of outbreaks and developing epidemics • Implementation of appropriate control measures • Surveillance studies.
There are four most common types of epidemiological studies:
• Cohort Study.
• Case Control Study.
• Occupational Epidemiological Study.
• Cross-Sectional Study.
• Epidemiologists use statistical methods to determine whether the differences they see are real or due to chance fluctuations.

16.Streaking Plates method?

Streak plate technique is used for the isolation into pure culture of the organisms (mostly bacteria), from mixed population. The inoculum is streaked over the agar surface in such a way that it “thins out” the bacteria. Some individual bacterial cells are separated and well spaced from each other.

17.Process of bacterial identification?

The identification of bacteria is a careful and systematic process that uses many different techniques to narrow down the types of bacteria that are present in an unknown bacterial culture. It produces benefits for many aspects of the research of microorganisms and helps physicians correctly treat patients.

18.Physical Method of Sterlization?

Heat Method of Sterilization

This is the most common method of sterilization. The heat used kills the microbes in the substance. ... In heatsterilization process, the longer the exposure to heat the better is the sterilization at a given temperature.

19.Disadvantage of Chlorine Sterilization?

disadvantages as well; First of all high concentration of this agent corrodes metals and also damages cloths. Secondly the strength of this solution decreases with time so whenever you want to use it always prepare its fresh solution.

20.Advantage of dry heat Sterilization?

Advantages of dry heat **sterilization** include: it is inexpensive, it does not pollute or cause toxic fumes, and it does not corrode or rust metal objects.**Disadvantages** include: it is relatively slow and many objects cannot withstand the high temperatures.

21.Virulence

The extent of pathogenicity is called virulence of a pathogen.

22.Name the three subgroups of spiral bacteria

These are curved shaped bacteria. They are further divided into 3 more subgroups.
a. Vibrio: curved rods
b. Spirillum: Helical but rigid
c. Spirochete: Helical but flexible
Spirochetes move by axial filaments which are enclosed by the outer membrane

23.flagellum and function (2

Flagellum: Long filamentous structure that propels bacteria It helps the bacteria in movement.

24.Name the Types of Bacteria on the basis of Oxygen Requirement

a. Obligate Aerobes: Do not grow without oxygen. It is necessary.
b. Facultative Anaerobes: Oxygen is necessary but start become anaerobes in its absence and can grow without of O_2.
c. Obligate Anaerobes: Cannot grow in the presence of oxygen.
d. Aerotolerant aerobes: Do not need O_2, also do not bother its presence
e. Microaerophiles: Aerobes but need small amount of oxygen. Cannot grow if available in large amount.

25.conjugation, transformation and transduction (3)

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Transformation
Naked DNA in solution is transferred from one bacterium to another during this process.
Conjugation
Genes transfer process can be mediated by plasmids. Plasmids are extra chromosomal circular DNA fragments that replicate independently of bacterial chromosome. Conjugation requires direct cell to cell contact and cells have to be
opposite mating types.
Transduction
Transferring of a gene from a bacterium to another bacterium via a virus is called transduction. o When a bacteriophage infects a bacterium such as E. coli, it replicates inside E. coli and also produces a protein coat (which is called a capsid) in which viral DNA is packed before the virus is released.

26.how disease can be transmitted (5)
Direct Transmission: person to person physical contact is a direct contact. Influenza spread is a typical example. Indirect Transmission: A nonliving object is involved. Objects in the use of a patient such as handkerchief, utensils, pillow and bedding are called fomites. These can transmit organisms to susceptible individuals.
Droplet Transmission: Mucus droplets are created when you sneeze for example. These droplets carry organisms that may infect another individual who comes in contact with the droplets. One sneeze may produce 20,000 droplets. Influenza can be spread by this route. Vehicle Transmission: Transmission of disease agents by a medium, such as water, food, or air. Other media include blood and other body fluids, drugs, and intravenous fluids.
What are Vectors? A vector is a vehicle (animate or inanimate) that carries a pathogen from one host to another.
Arthropods are vectors for many diseases. There are two ways these arthropods can transmit organisms to susceptible hosts:
Mechanical Transmission: In this type of transmission, organisms stick to the body parts of insects and get transmitted passively from one location to another.
Biological Transmission: Microbes multiply in the insect and is transmitted actively by insect bites from one individual to another

27.Any two microbial identification method (5)
Classical or Conventional Method
Serological testing
Nucleic acid based testing … plz See detail on 49 to 54 lessons on handouts

28.epidemiology and methods (5)
The study of where and when diseases occur and how they are transmitted in a population Descriptive Epidemiology: (recording the data about disease). It includes location and time of cases of disease, gender, health, age of patients etc. (Basically, frequency and distribution of risk factors in population are recorded).
Prevalence and incidence of disease are typical examples. Such studies can be retrospective (after they occur) or prospective (before they occur). So, in summary, in descriptive epidemiology, information about an already existing disease is recorded.
Analytical Epidemiology: It relates to determining the cause of the disease. Analysis of data, mode of transmission of diseases and means to prevent diseases also come in this discipline. Such studies can be done with case control methods in which factors that have preceded the disease are determined. A group of people who have the disease is compared with another group of people in the same location who are free of the disease. A Cohort method tracks two groups forward from exposure to outcome. Cohort study compares the experience of a group exposed to the factor with another group not exposed to the factor. An association between exposure and its outcome is determined which makes it easy to see the risk factor involved. In other words, cohort studies begin with a group of people (a cohort) free of disease. The people in the cohort are grouped by whether or not they are exposed to a potential cause of disease. The whole cohort is followed over time to see if the development of new cases of the disease (or other outcome) differs between the groups with and without exposure. Experimental Epidemiology: A hypothesis relating to a disease is tested in this approach. For example, smoking cause cancer can be tested by designing an experiment in mice in which they are exposed to certain levels of smoke over time and then the outcome is observed. Another example will be testing of a drug for prevention of a disease.

29.Magnetostome (5)
Inclusions of iron oxide
Surrounded by invaginations of plasma membrane
Present in G negative bacteria
Act like a magnet. Bacteria can stick to iron containing rocks for nutrition.
Decompose H_2O_2 which is toxic for cells.

30.Define immunity, difference between innate and adaptive immunity in Detail (10)
Immunity: Ability of the body to protect itself from harmful effects of diseases whereas Susceptibility is the lack of
immunity which provides the pathogens easy establishment to cause the diseases Types of Immunity
Innate (Natural):
Present at birth: One is born with these defenses, so it is present without any exposure to the microbes. That is the reason, it is non-specific.
Rapid to respond: Because it is already there, it interacts with the microbes as soon as they enter the body. Nonspecific: It means that it does not differentiate if it is an E. coli or a Staph or Salmonella. Body defense system against these entire pathogens acts in a similar fashion. In other words, the same macrophage can phagocytize E. coli, Staph or Salmonella at the same time (if these pathogens are present there).
Does Not Have Memory. Do not increase in responses to the repeated exposure to the same agent. Therefore, it has no memory.
First line of defense: Physical barriers (skin, mucus membrane) provide the first line of defense as they prevent the entry of the organisms by providing a physical barrier which if broken will let the microbes go inside. Second line of defense: Phagocytes and their secretions provide a second line of barriers for microbes. So, if microbes succeed in breaking the first barrier, macrophages, neutro-phils and such other cells phagocytize these microbes and kill them before these can establish infection in the body.
Adaptive (Acquired):
It is the third line of defense. It is acquired by experience and specific to an organism. Specific: It means that antibodies made against E. coli will destroy E. coli only, and will not do any harm to a Staph
for example
Have memory: It means if antibodies have once been made against E. coli, this exposure will be remembered by the body, and if a second encounter is made with the same type of E. coli again, the body would mount a heightened immune response against E. coli because it has memory cells developed in the body which upon second stimulation proliferate quickly and mount a greater defensive response.
Increases with repeated exposure: Because of memory, body's immune response increase with every repeated exposure. It has a limit also!
Slower to respond: For the first time exposure, the body's response is slower and takes a long time compared with the innate immunity which is immediate.
Adaptive Immune Response can further be divided into: Humoral immune response (antibody response by B cells or lymphocytes) and Cellular immune response (T cells are made which kills microbial harboring cells)

31.cell membrane function
a. Selective permeable barrier
b. Passive and Active Transport
c. Respiration in microbes (bacteria)
d. Photosynthesis in microbes
e. Lipid synthesis
f. Cell wall parts are transported by a molecule called bactoprenol which is present in the plasma membrane.
g. PM has many receptors in it with which various ligands can bind and initiate signals for gene expression etc.

32.Glycoclyx, its composition and function
It is the outermost layer of the cell wall, viscous and gelatinous in nature that surrounds the cells. It is composed of

polysaccharide and polypeptide or both. If glycocalyx is organized and firmly attached to the cell, it is called a capsule. – Plays important role in virulence (degree of pathogenicity) – Can also be a part of the vaccine against the bacteria to which it belongs.
Glycocalyx is also part of biofilms that bacteria make to attach to surfaces. Bacteria secrete extracellular polymeric substance (EPS) Biofilm protects cells within it. It also facilitates communication amongst cells If glycocalyx is loosely attached to the cell, it is called slime.

33. Differentiate between vertical and horizontal gene transfer and name three horizontal gene transfers

Vertical Gene Transfer:
This is a normal way of transferring genes from parents to offspring. This happens when a cell divides. Each daughter cell receives exactly what its parent cell has.
Horizontal Gene Transfer:
When genes are transferred from cells to cells within the same species, the process is called horizontal gene transfer.
This can happen between cells of the same species, or across different species of organisms. Horizontal gene transfer involves a donor and a recipient cell. The recipient cell then incorporates received DNA into its own genome and this genome becomes a recombinant molecule or recombinant DNA. The cell that has this recombinant DNA in it is called a recombinant cell. Three processes are known by which genes can be transferred horizontally from one cell to them other, and they include:

1. Transformation
2. Conjugation
3. Transduction

34.three types of passive movement

Diffusion Facilitated Difussion Osmosis

35. Economic benefits of fungi

Saccharomyces are used for alcoholic beverages under anaerobic conditions and if incubated aerobically, it is used in bread making because it metabolizes glucose to produce CO_2 (some ethanol is also produced but gets evaporated during baking) which expands the dough causing it to rise. This yeast is also extensively used for molecular biology work and vaccine production.
Vitamin C is obtained from Aspergillus niger.
Cellulase from Trichoderma

Antibiotics from many fungi and anticancer Taxol from Taxomyces are some other examples of useful fungal products.
Entomophaga (kills gypsy moth), hence is used as a biological pest control.
Mushrooms can be cooked and eaten as a source of proteins for humans.

36.Pili and its functions

These are hair-like structure composed of pilin, usually one to ten in number. Longer than fimbriae Used for attachment to:
Host cells
Bacteria
Used for DNA transfer from one bacterium to another during Conjugation (Sex pili)
Also function in twitching Motility
Gliding Motility is also the function of the pili.

37.physical nature of media 2

- Solid Media
- Liquid Media

38.germination of spore, 5

Exosporium: A thin delicate outermost covering of the spore
Spore coat: 2 nd layer underneath the exosporium. It is thick and composed of several protein layers. Resistant to chemicals
It contains enzymes for germination. Germination of spores into vegetative form occurs when environment becomes favorable for their growth. Cortex: It is the 3rd layer from outside in. It has peptidoglycan in it. Spore cell wall or core wall: Surrounds the protoplast or spore core Spore core: Contains nucleoid and ribosomes

39.Give two advantages of solid media in bactrial culturing.2

Solidification of the medium is done for purifying organisms from each other as solid medium provides surface for individual colonies to grow well separated from each other.
Secondly, the solid media are used to study the colony characteristics.

40.Types of microscopy 5

- Dark field microscopy
- Phase contrast microscopy
- Fluorescent microscopy
- Transmission electron microscopy
- Scanning electron microscope

41.Writes stages of respiration of glucose 3

Glycolysis: Oxidation of glucose to pyruvic acid
Krebs cycle: Oxidation of acetyl CoA to CO2
Electron Transport Chain: Coenzymes that carry electrons from Krebs cycle or glycolysis are oxidized to create ATP.

42. Differentiate between horizontal and vertical gene transfer.

- **Vertical Gene Transfer** o This is a normal way of transferring genes from parents to offspring. This happens when a cell divides. Each daughter cell receives exactly what its parent cell has.
- **Horizontal Gene Transfer**

genes are transferred from cells to cells within the same species, the process is called horizontal gene transfer.
This can happen between cells of the same species, or across different species of organisms. Horizontal gene transfer involves a donor and a recipient cell. The recipient cell then incorporates received DNA into its own genome and this genome becomes a recombinant molecule or recombinant DNA. The cell that has this

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recombinant DNA in it is called a recombinant cell. Three processes are known by which genes can be transferred horizontally from one cell to the other, and they include:
1. Transformation 2. Conjugation 3. Transduction

43. Define buffers.

o Buffers are used to maintain desired pH.
o Bacteria grown in lab produce acids. o Growth inhibited by acids. o Peptones and amino acids are used as buffers. o Phosphates are also used for buffers.

44. Define immunity, innate immunity and adaptive immunity.

• Immunity: Ability of the body to protect itself from harmful effects of diseases.
• Innate (Natural):
o Present at birth: One is born with these defenses, so it is present without any exposure to the microbes. That is the reason, it is non-specific. o Rapid to respond: Because it is already there, it interacts with the microbes as soon as they enter the body. o Nonspecific: It means that it does not differentiate if it is an E. coli or a Staph or Salmonella. Body defense system against these entire pathogens act in a similar fashion. In other words, the same macrophage can phagocytize E. coli, Staph or Salmonella at the same time (if these pathogens are present there). o Does not increase in response with repeated exposures to the same agent. Therefore, it has no memory.

o **First line of defense:** Physical barriers (skin, mucus membrane) provide the first line of defense as they prevent the entry of the organisms by providing a physical barrier which if broken will let the microbes go inside. o **Second line of defense:** Phagocytic cells and their secretions provide a second line of barriers for microbes. So, if microbes succeed in breaching or breaking the first barrier, macrophages, neutrophils and such other cells phagocytize these microbes and kill them before these can establish infection in the body. **Adaptive Immunity:** The **adaptive immune** system, also known as the acquired **immune** system or, more rarely, as the specific **immune** system, is a subsystem of the overall **immune**system that is composed of highly specialized, systemic cells and processes that eliminate pathogens or prevent their growth. **Adaptive (Acquired):** It is the third line of defense. Acquired by experience, specific to an organism.

45. What is epidemiology? Describe its tyes.

o The study of where and when diseases occur and how they are transmitted in a population. There are three basic types of investigations relating to epidemiology:
Descriptive Epidemiology
Analytical Epidemiology
Experimental Epidemiology:

46. Enlist microbial identification methods.

Classical or Conventional Method
Serological Methods
Nucleic Acid based Methods

47. Write economical uses of fungi.

Saccharomyces are used for alcoholic beverages under anaerobic conditions for alcohol production (fermentatively). However, if Saccharomyces are incubated aerobically (for bread making), they metabolize glucose to produce CO_2 (some ethanol is also produced but gets evaporated during baking) which expands the dough causing it to rise. This yeast is also extensively used for molecular biology work and vaccine production. Vitamin C is obtained from *Aspergillus niger*; cellulase from Trichoderma, antibiotics from many fungi and anticancer Taxol from Taxomyces are some other examples of useful fungal products. Entomophaga (kills gypsy moth), hence is used as a biological pest control. Mushrooms can be cooked and eaten as a source of proteins for humans.

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48.What is sporulation?

• Formation of spores takes place within a vegetative cell and the process is called sporulation or sporogenesis. It is initiated when nutrients become unavailable.

50. What is flagellum? Write its functions.

Flagellum: Long filamentous structure that propels bacteria

Vertical Gene Transfer
Horizontal Gene Transfer
1. Transformation 2. Conjugation 3. Transduction

52. Describe classification system proposed by Robert H Whittaker.

In 1969, five kingdom classification was proposed by Robert Whittaker as under:
•Plantae: plants
•Animalia: Animals
•Fungi: Yeasts, molds and mushrooms
•Protista: These are unicellular eukaryotes. Organisms that do not fit into any other category are placed in Protista. They are larger than prokaryotes. They include algae, protozoa, slime molds and water molds.
•Prokaryotae or Monera: Bacteria

53. What is taxonomic hierarchy?

All organisms can be grouped into a series of subdivisions that make up the taxonomic hierarchy.
Specie:
A bacterial **species** represents ―a monophyletic and genomically coherent cluster of individual organisms that show a high degree of overall similarity with respect to many independent characteristics, and is diagnosable by a discriminative phenotypic property‖
Genus
Order
Family
Group
Class
Phylum

Kingdom
Domain

54. What is western blotting?

Western Blotting: We can use this technique for the detection of antigen in the serum. Microbial proteins can be separated on SDS-PAGE by electrophoresis and the presence of these proteins can be detected by enzyme-tagged antibodies specific to those proteins. A color band will be seen where the specific protein (antigen) is present on the gel. Please remember that proteins in the gel are first transferred to a paper strip before they could be detected by specific antibodies

55. Enlist types of passive transport.

Simple Diffusion
Facilitated Diffusion

Osmosis

56. Name types of spiral bacteria.

These are curved shaped bacteria. They are further divided into 3 more subgroups.
Vibrio: curved rods
Spirillum: Helical but **rigid**
Spirochete: Helical but **flexible**

57. Define passive movement.

Passive Movement
This means that substances will move from
higher concentration of a substance to lower concentration. Gradient provides the force for movements of molecules and
this happens with no energy expenditures.

58. What is function of lysozymes?

A lysosome is a membrane-bound organelle found in nearly all animal cells. They are spherical vesicles that contain hydrolytic enzymes that can break down many kinds of biomolecules. Lysosomes are organelles that contain digestive enzymes. They digest excess or worn out organelles, food particles, and engulfed viruses or bacteria.

59. What are fimbriae? Give location and function.

• Hair-like small appendages on G around the entire cell epithelial cells before they can secrete biofilm.
• Fimbriae also help attach to epithelial cells e.g *Neisseria gonorrhoeae*

60. Name types of culture medium on the basis of function.

a. Differential medium
b. Selective medium
c. Enriched medium
d. Basal medium
e. Reducing medium
f. Selective medium
g. General purpose medium

61. Write functions of cell wall, flagella and ribosomes.

Cell wall gives support and protection against plasmolysis.
Flagella help bacteria in movement.
Ribosomes are organelles that assist in protein synthesis during translation.

62. What are types of microscopes?

a. Light microscope
b. Electron microscope
c. Scanning electron microscope
d. Transmission electron microscope
e. Dark field microscope
f. Bright field microscope
g. Fluorescent microscope

63. What do you know about types of techoic acid?

G-positive cell wall contains teichoic acid which may be divided into wall teichoic (remains within the cell wall) acid and lipoteichoic acid (is inserted into the cell membrane).

64. Write characteristics of integral proteins.

i. Integral proteins act as channels or carriers in facilitated diffusion. Integral proteins are called transporters or permeases
ii. No energy is required during transportation of substances through these integral proteins.
iii. Two kinds of transporters are known:
1. Ions
i. Nonspecific transporters
ii. specific transporters
1. Change in shape
2. Sugars, vitamins

65. What are benefits of solid medium?

Solidification of the medium is done for purifying organisms from each other as solid medium provides surface for individual colonies to grow well separated from each other. A well isolated colony is assumed to be the progeny of a single cell. Secondly, the solid media are used to study the colony characteristics.

66. Define refractive index.

Refractive Index: Ability of a medium to bend the light o Light rays move in a straight line through a single medium o Light rays bend when the pass from one medium to another

67. Define medical microbiology.

Medical microbiology is a branch of **medical** science concerned with the prevention, diagnosis and treatment of infectious diseases. In addition, this field of science studies various **clinical** applications of microbes for the improvement
of health.

68. What is resolution?

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Resolution is the ability of the lenses to distinguish between two closely lying objects as separate.of Light microscope resolving power is 0.2 μm.

Prokaryotes
• DNA is not enclosed in a nuclear membrane.
• Chromosome: mostly Single, circular
• DNA not associated with histones
• No membrane enclosed organelles
• Cell wall has peptidoglycan, a complex carbohydrate.
• Divide by binary fission.

Eukaryotes
• DNA is enclosed in a membrane-bound nucleus.
• DNA is found in multiple chromosomes.
• Chromosomes are linear, thread-like structures.
• DNA is associated with histones.

Cell wall if present is chemically simple cellulose or chitin
they have membrane enclosed organelles **mitochondria chloroplasts endoplasmic reticulum golgi apparatus lysosomes etc.**

70. Define virulence. Virulence:
The extent of pathogenicity

1. Plate Counts
a. Pour Plate Method:
b. Spread Plate Method:
2. Filtration:
3. The most Probable Number Method:
4. Direct Microscopic Count

72.Write steps of germination of spores.

• Spore germinates when it finds a conducive environment for its growth. Germination has 3 phases or stages:
Activation: It prepares the spore for germination. conment.
Germination: Spore starts swelling and loosing its coats etc.
Outgrowth: New components are made.

73. Write types of lipopolysaccharides. 2

74. Define culture,

Microbial growth in the lab is called a **culture**.
A nutrient material that supports the growth of microbes in the lab is called a **culture medium**.
Microbes introduced into a culture medium that initiate growth of organisms are known as **inoculum.**

75. Define antiseptics.

Compounds that are used to remove pathogens from living tissue are known as antiseptics.

76. Define phosphorylation.

The addition of an inorganic phosphate group to a chemical compound is called phosphorylation. e.g conversion of ADP into ATP in living tissues.

77. Explain phases of growth curve.

In a batch culture, organisms go through various phases of growth. If we follow this growth with respect to time, andmake a graph of this growth pattern, it will give us what we call as a growth curve. There are four distinct phases of this curve. **1-Lag phase 2- Log phase 3- stationary phase 4- Death phase.**

78. What is high frequency of combination?

In some cells that carry F_+, F_+ (Fertility factor) gets incorporated in the chromosomal DNA which converts F_+ cells to high frequency of recombination cells (Hfr cells). When conjugation occurs between an Hfr cell and an F_- cell, the Hfr cell's chromosome (with its integrated F factor) replicates, and a parental strand of the chromosome is transferred to the recipient cell. Since this transfer starts from the middle of F factor gene, and most of the time, this transfer is not complete; the whole chromosome of Hfr cell is not transferred. However, some genes can be transferred during this process. When these genes become integrated into the genomic DNA of the recipient cells, the recipient cells acquires new versions of genes that were not part of its genome previously. It may be noted that F_- cells remains F_- negative cell because F factor is not transferred completely. This whole process is illustrated in the accompanying diagram.These newly acquired genes can be mapped easily with respect to the time they get transferred to F_- cells.
Clone: Population of cells derived from a single cell that are genetically identical.
Strain: A genetic variant of a clone is called a strain.

79. Differentiate between clone and strain?

Clone: Population of cells derived from a single cell that are genetically identical.
Strain: A genetic variant of a clone is called a strain.

80. Differentiate between positive sense and negative sense RNA.

Zoonosis: Primarily an animal infection, but can be transmitted to humans. It can assume epidemic proportions in humans and is difficult to control or eradicate.

82. What are interferons? Give their function.

Interferons are antiviral proteins, produced by macrophages and lymphocytes in response to viral infections. Fibroblasts can also secrete it. These proteins interfere with viral replication. Thereare two types of interferons:
Species specific: This means that interferon produced by humans cannot be used in dogs or cats or

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vice versa.
Not viral specific: This means that interferons produced in response to virus A can also be effective against virus B.

83. Explain characteristics of aerophiles.

84. Describe conjugation, transduction and transformation. Horizontal gene 10

transfer:

When genes are transferred from cells to cells within the same species, the process is called horizontal gene transfer. This can happen between cells of the same species, or across different species of organisms. Horizontal gene transfer involves a donor and a recipient cell. The recipient cell then incorporates received DNA into its own genome and this genome becomes a recombinant molecule or recombinant DNA. The cell that has this recombinant DNA in it is called a recombinant cell. Three processes are known by which genes can be transferred horizontally from one cell to the other, and they include:
1. Transformation 2. Conjugation 3. Transduction **1. Transformation** o Naked DNA in solution is transferred from one bacterium to another during this process. Frederick Griffith observed this process in 1928 when he was working with *Strep pneumonia* in his lab. This bacterium has two strains: one is capsulated and is called a Smooth strain (which is pathogenic) and the other is Rough strain (non-pathogenic).

o When live encapsulated bacteria are injected into a mouse, the mouse gets killed by the microbe. When rough strain is injected, the mouse is not killed. Moreover, when heat-killed encapsulated bacteria are injected, it also does not kill the mouse. However, when heat-killed strain is mixed with rough live strain, and a mouse is injected with this mixture, the mouse dies. If you isolate the bacterium from these dead mice, the bacteria resemble with encapsulated strain of *Strep pneumonia.* The interpretation of this experiment led to the mechanism that rough live bacteria take up DNA from the heat-killed encapsulated strain and becomes pathogenic by acquiring the ability to form a capsule. Very few bacteria have this property of accepting foreign DNA and most of it is degraded (only small fragments of DNA get incorporated in the genome of the recipient. The cells that can take up DNA are called competent cells. We can create such microbes in the lab that can accept a given DNA during gene transfers that we undertake in our DNA work. The cells that take up the DNA and incorporate this in their genome successfully are called transformed cells.

2. **Conjugation** o Genes transfer process can be mediated by plasmids. Plasmids are extra- chromosomal circular DNA fragments that replicate independently of bacterial chromosome. In other words, a given bacterial cell can have multiple copies of plasmid with the same cell. Plasmids often have genes that are not strictly needed by the microbe for its growth. However, the genes on the plasmids provide some advantages to the bacterium. For example, antibiotic resistant genes offer a mechanism for the bacterium to resist the presence of that antibiotic. In other words, bacterium that has a plasmid with an antibiotic gene on it can still multiply in the presence of that antibiotic.

o Conjugation requires direct cell to cell contact and cells have to be opposite mating types. Plasmids in G –ive cells have genes that code for sex pili because pili are needed to make a bridge between the two mating type cells. Since there is no pilus present in G +ve bacteria, they secrete sticky surface molecules instead which create contact between the two cells. The pilus along with two mating cells is shown in the diagram below:

o These plasmids are transmissible from one bacterium to another during conjugation.

o Plasmid is replicated and a single copy of the plasmid is transferred to the recipient bacterium through the pilus.

A complementary copy is then made in both the cells to make the DNA of the plasmid double stranded molecule.
o In E. coli, fertility factor (F factor) plasmid was the first plasmid observed to be transferred from one organism to the other, hence those bacteria that have this plasmid are called F^+ cells to differentiate from those that do not have one (F^-). However, once F^- cells acquire F^+ plasmid, they also become F^+.

3. Transduction

o Transferring of a gene from a bacterium to another bacterium via a virus is called transduction. o When a bacteriophage infects a bacterium such as E. coli, it replicates inside E. coli and also produces a protein coat (which is called a capsid) in which viral DNA is packed before the virus is released. What happens is that when phages cause lysis of infected cells, bacterial chromosomal DNA also gets fragmented and such fragments can sometimes be packed into the viral capsids much like the viral DNA. Now, the capsid carrying the bacterial gene or genes in that fragment can infect another E. coli and can inject its DNA (bacterial DNA fragment) into it resulting in the transfer of genes from one bacterium to another.

85. Explain methods of microbial control in detail.

These methods can be broadly classified into four groups: Physical, Mechanical, Chemical and Biological

Physical methods of microbial control

Heat: It is the cheapest sources of all and easily available to control microbial growth. There are a few heat related concepts that we must appreciate. Heat denatures enzymes.
Thermal Death Point: It is the lowest temperature at which all cells in a culture are killed in 10 min.
It will be a specific temperature for a specific species of organism.
Thermal Death Time: Time (minimum) during which all cells in a culture are killed at a given temperature. This will vary from temperature to temperature for the same organism. Obviously, higher temperatures will take less time to kill the organisms than low temperatures. rther be classified into **dry and moist** heat methods.
Heat kills by oxidation
Form of dry heat include:
1. Dry heat (hot-air oven),
2. Flaming (Platinum loop sterilization in the flame of Bunsen burner),
3. Incineration (burning to ashes).
^{o}C, and Tyndallization (100°C for 3 consecutive days to sterilize sugar solution that can be degraded by autoclaving) and pasteurization (usually at low temperature than 100°C).
Pasteurization: This technique typically employs low heat for killing pathogenic and food spoiling bacteria in milk. However, it does not kill thermoduric organisms. Thermoduric organisms are not pathogenic though. Products other than milk, such as ice cream, yogurt, and beer, all have their own pasteurization times and temperatures, which often di er considerably. Dairy industry tests phosphatase to determine if pasteurization of dairy product has occurred effectively because phosphatase present in raw milk gets inactivated with effective pasteurization temperatures. The following three equivalent heat treatments can be given to achieve pasteurization of milk:
o 63°C for 30 min
o o High-temperature short-time: 72°C for 15 sec
o Ultra-high-temperature: 140°C for 4 sec

Other Physical Methods

Low temperature inhibits microbial growth by slowing down chemical reactions.
Refrigeration: Materials stored at 2-8°C can last for a day or two.

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Deep-freezing: Long term storage of materials is possible at minus temperatures in the deep freezers.
Lyophilization: Freeze drying is another method for long term storage of food and other materials.
High pressure deshapes and denatures proteins: This technique is used for preserving juices and wines.
Desiccation :prevents metabolism: This method is very old and still in use today. Food stored in high concentrations of salt or sugar solution can keep for long.
Osmotic pressure causes plasmolysis. High salt concentrations for example can take the water content of the cells out leaving them starved of water.
Filtration: It is also a physical method. It can be used to filter air (using high efficiency particulate air filter) or liquid medium using membrane filters.
Radiation: o Ionizing and non-ionizing radiation o Ionizing Radiation: High energy waves that destroys microbes.
• Ionization radiations include gamma rays, X rays, or high-energy electron beams. The ionizing radiation possesses a wavelength shorter than that of nonionizing radiation, less than about 1 nm. All these short wavelengths cause ionization of water generating high reactive hydroxyl radicals which are damaging to the cells as they bind to DNA and results in mutations. Medical plastic supplies, medicines and meat products can be sterilized by radiation. -ionizing Radiation: wavelengths longer than 1nm fall into non-ionizing radiation. The best example is UV light. UV light causes mutations in the DNA by dimerization of thymidine bases. These *thymine dimers* inhibit correct replication of the DNA during reproduction of the cell. The most useful wavelength is 260 nm. This wavelength is specifically absorbed by DNA. UV radiation is used to control organisms in the air mostly. However, surfaces are also sterilized by UV radiation. UV light cannot penetrate deep into the fluids, so only surface bacteria are killed.

ETHYLENE OXIDE FOR MICROBIAL CONTROL

o **Ethylene Oxide:** Many heat-sensitive items such as disposable plastic, petri dishes and syringes, heart-lung machine components, sutures, and catheters are sterilized by ethylene oxide. Ethylene oxide is a strong alkylating agent that kills by reacting with functional groups of DNA and proteins to block replication and enzymatic activity. It rapidly penetrates packing materials, even plastic wraps. EtO is explosive, supplied in a 10% to 20% concentration mixed with either CO_2 or dichlorodifluoromethane. A clean object can be sterilized if treated for 5 to 8 hours at 38°C or 3 to 4 hours at 54°C when the relative humidity is maintained at 40 to 50% and the EtO concentration at 700 mg/L.

CHEMICAL METHODS FOR MICROBIAL CONTROL

Chemicals are used to control microbial growth. Factors that influence the efficacy of a chemical disinfectant include the concentration of the disinfectant, presence of organic matter in the environment (matrix) where the disinfectant is being used, pH of the environment (disinfectants are more effective at acidic pH), and time of exposure to the disinfectant.

86. How infectious diseases are transmitted? Explain in detail.

1) how infectious disease can be transmitted explain it briefly transformission occur to contact ,vehicle n vector.(10)

Direct Transmission: person to person physical contact is a direct contact. Influenza spread is a typical example.
Indirect Transmission: A nonliving object is involved. Objects in the use of a patient such as handkerchief, utensils, pillow and bedding are called fomites. These can transmit organisms to susceptible individuals.
Droplet Transmission: Mucus droplets are created when you sneeze for example.
These droplets carry organisms that may infect another individual who comes in contact with the droplets. One sneeze may produce 20,000 droplets. Influenza can be spread by this route.
Vehicle Transmission: Transmission of disease agents by a medium, such as water, food, or air. Other media include blood and other body fluids, drugs, and intravenous fluids.
Vectors o A vector is a vehicle (animate or inanimate) that carries a pathogen from one host to another. O Arthropods are vectors for many diseases. There are two ways these arthropods can transmit organisms to susceptible hosts:
Mechanical Transmission: In this type of transmission, organisms stick to the body parts of insects and get transmitted passively from one location to another.
Biological Transmission: Microbes multiply in the insect and is transmitted actively by insect bites from one individual to another.
Nosocomial Infections o Hospital acquired infections are called nosocomial infections.

87. Explain salient features of viruses, virions and viroids.

VIROIDS

These are infectious agents that consist only of RNA and found in plants only. No capsids, no envelopes, just RNA. They are covalently closed circular ssRNAs about 250 nucleotides long. The circular RNA normally exists as a rodlike shape due to intrastrand base pairing, which forms doublestranded regions with single-stranded loops. They do not encode any gene products, so they are replicated by cellular DNA dependent RNA polymerase. A plant infected with such viroids may not show any signs of illness. However, in other plants, they may cause severe disease. They cause the disease by RNA silencing process.

VIRUSOIDS

o They are similar to viroids; however, they encode one or two gene products and need a helper virus to infect the host cells. The helper virus supplies gene products and other materials needed by the virusoid for completion of its replication cycle. Hepatitis D virusoid is a typical example which uses Hepatitis B virus as a helper.

PRIONS

o These are proteinaceous infectious particles that cause neurodegenerative diseases such as scrapie in sheep, bovine spongiform encephlopthy, and Kuru in humans. These are abnormal form of cellular proteins. How these proteins accumulate in the cell is not clear, although, some genetic components are known to be involved.

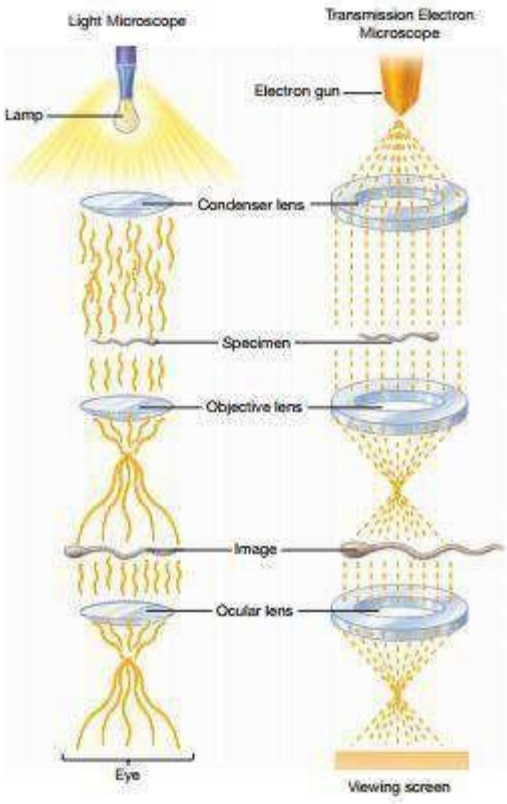
89. How fungi can be identified by biochemical testing?

Yeast if grown on solid medium grows like bacteria. In other words, colonies of yeast look like bacterial colonies but are bigger in size.
carbohydrates aerobically and produces CO_2 and water; however, if oxygen is denied, carbohydrates get converted fermentatively into ethanol and CO_2.

90. IDENTIFICATION OF SELECTED FUNGI

i are cultured using a slide (slide culture) on top of Sabouraud's agar. Spores are placed on top of the medium or embedded in the medium. A block of the medium can be used to grow fungi or a glass coverslip can be used to grow spores into hyphae and mycelium and spore arrangements can be studied under the microscope by staining with lactophenol cotton blue. Slides stained with lactophenol cotton blue can also be seen under the microscope.Stained slides can show specific arrangements of spores, their sizes and other morphological features as seen in the figure for Aspergillus.

91. Label the diagram of Electron microscope with numbers mentioned.



92. write functions of cell membrane of bacteria

- Selective permeable barrier
- Passive and Active Transport
- Respiration in microbes (bacteria)
- Photosynthesis in microbes
- Lipid synthesis
- Cell wall parts are transported by a molecule called bactoprenol which is present in the plasma membrane.
- PM has many receptors in it with which various ligands can bind and initiate signals for gene expression etc.

93. Write names of 5 types of bacteria with respect to temperature.

Organisms are basically classified into three groups based on the temp requirements:
1. **Psychrophiles:** These are further divided into strict psychrophiles and psychrotrophs:
o **Psychrotrophs:** Cold loving: 15 ^{0}C
o **Psychrotrophs:** Optimum temp is 20-30 ^{0}C. Food spoilage bacteria that can spoil food during refrigeration.
2. **Mesophiles :** 25 – 40 ^{0}C: These are the ones that cause diseases in animals and humans.
o Moderate temp loving organisms o Optimum: 37 ^{0}C as this is the body temperature of humans and animals.
3. **Thermophiles:** 50 – 60 ^{0}C
o Heat loving o Important in organic compost piles o These are further divided into hyperthermophiles that grow optimally at 80 ^{0}C. They live in hot springs. Every group of these organisms has a:
o Minimum growth temp o Optimum growth temp. o Maximum growth temp.

94. Write characteristics of metachromatic granules.

Metachromatic Granules

- Also called volutin, they stain red with certain dyes such as methylene blue. That is why they are called metachromatic (stain in different color as methylene blue gives blue color but the color on these granules is red).
- Large inclusions
- These inclusions contain inorganic phosphates. Inorganic phosphates are used up in ATP synthesis.
- Characteristics of *Corynebacterium diphtheria: This bacterium can be identified by the presence of these granules in it.*

95. Explain five kingdom classification systems.

In 1969, five kingdom classification was proposed by Robert Whittaker as under:
•Plantae: plants
•Animalia: Animals
•Fungi: Yeasts, molds and mushrooms
•Protista: These are unicellular eukaryotes. Organisms that do not fit into any other category are placed in Protista. They are larger than prokaryotes. They include algae, protozoa, slime molds and water molds.
Prokaryotae or Monera: Bacteria and Archea

96.Explain layers of endospore.

Endospore and its various parts/structures

- **Exosporium:** A thin delicate outermost covering of the spore
- **Spore coat:** 2nd layer underneath the exosporium. It is thick and composed of several protein layers Resistant to chemicals It contains enzymes for germination. Germination of spores into vegetative form occurs when environment becomes favorable for their growth.
- **Cortex:** It is the 3rd layer from outside in. It has peptidoglycan in it.
- **Spore cell wall or core wall:** Surrounds the protoplast or spore core
- **Spore core:** Contains nucleoid and ribosomes

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97. Enlist types of bacteria on the basis of oxygen.

Based on the use and concentration of Oxygen, microbes can be classified into five groups:

Obligate Aerobes: Oxygen must be present for their growth. These organisms do not grow if O_2 is not present in the medium. If you test these organisms in a tube containing nutrient agar, the growth will remain confined to the surface only (see the diagram for the concept). Mycobacterium tuberculosis is an example of such an organism.

Facultative Anaerobes:

These organisms prefer aerobic respiration; however, if oxygen is not available, they can use anaerobic respiration or fermentative modes for generating ATPs. These organisms will be present throughout the tube, however, more growth will be seen on the top (close to the surface) because these organisms prefer using oxygen if it is available.

Obligate Anaerobes:

Although, these organisms do not use oxygen, they cannot grow in the presence of oxygen. The reason is that oxygen creates toxic compounds which can kill these cells. Normally, oxygen toxic compounds are eliminated by the cells, however, strict anaerobe do not have a system to get rid of these toxic compounds when they are made in the cells. So using or not using oxygen is one thing and growing in the presence of oxygen (and not utilizing it) is

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o **Aerotolerant Aerobes:** These organisms do not use oxygen and also are not bothered by the presence of oxygen. In other words, they have a better system to dispose of toxic oxygen compounds. They will grow evenly throughout the tube of medium\

98.Name five types of microscopy . 5 marks

Dark field microscopy

o This microscope makes use of a special condenser that provides illumination from the sides only as depicted in the diagram below:

o This special arrangement of the condenser throws light rays at an oblique angle so that if the specimen is not present on the slide, there will be no image formed. However, if the the organisms are present on the glass slide, those organisms will reflect the light into the objective lens and we will see the cells as bright objects in a dark background much like we see stars in the sky at night.

o The beauty of this microscope is that one does not have to stain the organisms to see them under the microscope.

Phase contrast microscopy

This instrument is built on the concept of phases of light waves. The light ray that passes through an organism gets retarded about ¼ of its wavelengths than the

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light that passes outside the organism (through the glass slide only). To further retard the retarded light ray, a phase plate is inserted into the body of the microscope (above the objective lens) which is thicker in the center than the ends. This arrangement causes the retarded light to further slow down about ¼ of its wavelength, which when recombines with the un-retarded light creates a contrast which makes the cells/organisms visible.
o This method for visualizing organisms also does not involve staining.

Fluorescent microscopy

o This microscope is equipped with UV light that provides the source of light for illumination. However, this light source does not pass through the slide, it instead falls over the slide (having specimen) and creates fluorescence.
o Fluorochromes are substances that absorb short wavelength of light and emit longer wavelength (visible light spectrum)
o Fluorochromes are attached to antibodies that are specific to organisms. So, if an organism is present in the specimen, we can use specific antibodies to determine the presence of those organisms by use of fluorochrome tagged antibodies. See the accompanying diagrams for details. Look for UV lamp (Mercury Lamp).

Transmission electron microscope

o Electron microscope (EM) makes use of electron beam as an alternative to light rays. Since electron beam has much shorter wavelength, resolution of EM is 0.1nm compared with light microscope which is 200nm.
o Second difference of EM with light microscope is that focusing of electron beam is done not by glass lenses but by electromagnets that acts like lenses.
o The third difference in EM is that the image of the specimen is formed on a fluorescent screen like a TV screen.

Scanning electron microscope

This is a modified version of electron microscope. Electron beam is allowed to fall on to the specimen and the image is formed from the reflected electrons that are collected and assembled into an image by the detector. It gives a three dimensional picture.

99. Explain ELISA.

There are some interactions (Antigens and Antibodies) that are not directly evident. Interactions may have taken place, but there is no visible direct clue if the interaction has taken place or not. For such interactions, we use indirect methods. Say, there is big body of smoke visible to you from a kilometer ahead of you. What is your guess? Yes, you guessed it right. There is a big fire there. Did you see the fire? Probably not. You guessed on the basis of smoke that there is a fire there. This is called indirect way of interpretations. A beautiful example of such a test is ELISA where we do not know if the interaction has taken place or not. However, we can see it indirectly. We will see that in the following paragraph which discusses ELISA.

Enzyme-Linked Immunosorbent Assay:

This test is done by coating the ELISA plastic plates either with the antigen or the antibodies. Suppose we have Brucella antigen coated ELISA plates that can detect antibodies from the serum of a Brucella infected animal. You take the serum and put into the wells of ELISA plate. You allow time for antigen and antibody to interact and then wash these wells with some buffer to remove unbound antibodies. These bound antibodies are not visible at this step. So, how do we see this interaction? Well, we use another antibody that has been made against the antibody bound to the antigen. This special antibody has been tagged with an enzyme. After allowing time for another. The growth will remain confined to the bottom which has little oxygen.
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This is a modified version of electron microscope. Electron beam is allowed to fall on to the specimen and the image is formed from the reflected electrons that are collected and assembled into an image by the detector. It gives a three dimensional picture.

99. Explain ELISA.

There are some interactions (Antigens and Antibodies) that are not directly evident. Interactions may have taken place, but there is no visible direct clue if the interaction has taken place or not. For such interactions, we use indirect methods. Say, there is big body of smoke visible to you from a kilometer ahead of you. What is your guess? Yes, you guessed it right. There is a big fire there. Did you see the fire? Probably not. You guessed on the basis of smoke that there is a fire there. This is called indirect way of interpretations. A beautiful example of such a test is ELISA where we do not know if the interaction has taken place or not. However, we can see it indirectly. We will see that in the following paragraph which discusses ELISA.

Enzyme-Linked Immunosorbent Assay:

This test is done by coating the ELISA plastic plates either with the antigenor the antibodies. Suppose we have Brucella antigen coated ELISA plates that can detect antibodies from the serum of a Brucella infected animal. You take the serum and put into the wells of ELISA plate. You allow time for antigen and antibody to interact and then wash these wells with some buffer to remove unbound antibodies. These bound antibodies are not visible at this step. So, how do we see this interaction? Well, we use another antibody that has been made against the antibody bound to the antigen. This special antibody has been tagged with an enzyme. After allowing time for interaction of this 2nd antibody with the 1st antibody, 2nd antibody is also given a washing to remove unbound antibody molecules. Again, we cannot say if interaction has taken place or not. Here comes an indirect way of detecting this interaction. Remember that the 2nd antibody has a molecule of an enzyme attached to it. Now, if we provide substrate to the enzyme in the wells, enzyme will convert the substrate into a color compound visible to the eyes. The color change can be measured by a spectrophotometer as well which gives a quantitative data

100. Explain types of algae.

o **Phaeophyta (Kelp):**

Commonly known as brown algae, brownish in color, contains cellulose and alginic acid cellwalls, are multicellular and contain chlorophyll *a* and *c*, xanthophylls. These store carbohydrates and are used for algin which is a thickener in ice creams. Can grow upto 20 cm in one day.

o **Rhodophyta:**

Red Alage, these are reddish in color and contain cellulose in their cell walls. Most are multicellular and contain chlorophyll a and d, and phycobiliproteins. They store glucose polymer and used for agar and carrageenan (both are used as thickeners of foods).

o **Chlorophyta:**

Green Algae: have all the features of red algae except that chlorophyll is of a and b types only. These algae are believed to have given rise to terrestrial plants.

o **Bacillariophyta:**

Cell wall is composed of pectin and silica. They are unicellular and possess chlorophyll a and c, and carotene xanthophylls. These algae store oil and produce domoic acid with which human can be intoxicated by eating contaminated mussels. Diatoms: Come in beautiful colors and shapes.

o **Dinoflagellata:**

These are unicellular algae collectively called as plankton (free floating organisms). These cells have cellulose in their cytoplasmic membrane which gives the membrane rigidity. They have chlorophyll *a* and *c*, and carotene. These algae store starch in them. Dinoflagellates in the genus Alexandrium produce neurotoxins (called saxitoxins) which can cause paralytic shellfish poisoning in humans when they consume contaminated mussels and clams. Blooms of Alexndrium gives ocean a deep red color commonly known as red tide.

Oomycota (Water molds):

They are basically decomposers of dead matter. Superficially, these resemble with fungi; however, these are motile with flagella and their cell wall is composed of cellulose not chitin. Their spores are called zoospores.

o Terrestrial members of water molds are also plant parasite. *Phytophthora infestans* was responsible for Irish potato blight during mid-1800s.

P. ramorum causes sudden oak death

101. Define magnification

Magnification is the enlargement of an object its formula is Magnification of objective lens x magnification of ocular lens

102. Define Buffer

A buffer is the substance that maintains the desired pH in the solution or any medium.

103. What is taxonomic hierarchy

Microbes are placed in groups based on similarities that they share with each other. All organisms can be grouped into a series of subdivisions that make up the taxonomic hierarchy.

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104. Give two advantages of solid media in bacterial culturing.

a. Pure culture is easy to obtain by streaking the organisms
b. Colony characters can be studied

105. Physical Requirements of Media

Physical: Requirements • Temperature • pH • Osmotic Pressure

106. Flagellum and its function (2)

Long filamentous structure that pels ba the primary role of the flagellum is locomotion, but it also often has function as a sens y anelle, eing sensitive to chemicals and temperatures outside the cell.

Example: Chemotaxis —Phototaxis.

107. Glycoclyx, its composition and function

It is the outer I layer, is and gelatinous in nature that surrounds the cells. It is composed of polysaccharide and pipe or h. • Capsule: If glycol calyx is organized and firmly attached to the cell, it is called a capsule.— Plays important role in virulence (degree of pathogenicity)—Can also be a part of the vaccine against the bacteria to whi it b n an be demonstrated by negative staining technique.

108. Charcteristic of microaerophiles

A mi aerophile is a microorganism that requires oxygen to survive, but requires environments containing lower levels of oxygen than are present in the atmosphere (i.e. <21% 0,; typically 2—10% 0,). Many microaerophiles are also capnophiles, requiring an elevated concentration of carbon dioxide (e.g. 10% CO, in the case of *Campylobacter* species.

109. Write composition of teichoic Acid?

Teichoic Acid Composition: - (glycerol or ribitol + Phosphate)

110. Differ b/w chemotaxis and phototaxis

Phototaxis: -The movement of organism or its part towards the light while s the movement of an organism towards specific stimulus or chemical substance called chemotaxis.

What is western blotting

We can use this technique for the detection of antigen in the serum. Microbial proteins can be separated on SDS-

W

PAGE by electrophoresis and the presence of these proteins can be detected by enzyme-tagged antibodies specific to those proteins

111. Define virulence

The ability of an agent of infection to produce disease, the **virulence** of a microorganism is a measure of the severity of the disease it causes.

112. Two teichoic acid present in bacterial cell wall.

1) Lipoteichoic Acid 2) Wallteichoic acid

113. Define antiseptics

An **antiseptic** is a substance that stops or slows down the growth of microorganisms. Like Ethanol , hydrogen peroxide etc.

114. What is high frequency combinations

If F factor gets integrated into the chromosome of the recipient cells, the bacterium becomes high frequency of recombination cell, it is a kind of conjugation.

115. Differentiate between clone and strain

Clone: Population of cells derived from a single cell that is genetically identical. **Strain:** A genetic variant of a clone is called a strain

116. Difference between positive sense and negative sense RNA

The main difference between positive and negative sense RNA virus is that positive sense RNA virus consists of viral mRNA that can be directly translated into proteins whereas negative sense RNA virus consists of viral RNA that is complementary to the viral mRNA.

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117. Difference between positive sense and negative sense RNA

"The main difference between positive and negative sense RNA virus is that positive sense RNA virus consists of viral mRNA that can be directly translated into proteins whereas negative sense RNA virus consists of viral RNA that is complementary to the viral mRNA.

118. Name two basic dyes

Crystal violet • Methylene blue • Malachite green • Safranin

119. Two methods for Viable counting

1) Spread plate 2) Pour plate

120. what is microbiology

Studying microbes microbiology

121. characteristics of Integral proteins.

Imbedded in the membrane —Insoluble in water —Amphipathic in nature

122. Formula of magnification?

Magnification of objective lens x magnification of ocular lens

123. Substrate level phosphorylation

The Direct transfer of P to ADP

124. Macconky agar

Nutrient agar + Bile salt + CV • Selective & differential, Used for gut bacter"

125. what is virulence?

Virulence: The ability of an agent of infection to produce disease. The *virulence* of a microorganism is a measure of the severity of the disease it causes.

126. Advantages of solid medium.

Pure culture is easy to obtain by streaking the organism ” • Colony characters can be studied.

127. Pasturization method

It is a method in which the liquid usually milk a rized Reduces spoilage organisms and pathogens

128. Two types of fermentation?

Two types are follows.

- ✓ Lactic acid fermentation
- ✓ alcoholic fermentation

129. define L-form bacteria 2m

L-form organisms are naturally found without the cell wall. In other words, normally, they have cell wall but under certain conditions, they hacve cell wall but under certain condition . gthey may lack a cell wall.

130.. Industrial microbiology

Use of microbes or their enzymes for large scale production of biomolecules

131. Describe transduction , conjugation transformation

a. **Transduction:-** Transferring of a gene from a bacterium to another bacterium via a virus is called transduction.

b. **Conjugation:-** Bacterial conjugation is the transfer of genetic material between bacterial cells direct cell- to-cell contact or by a bridge-like connection between two cells.

c. **Transformation:-** Transformation is the process by which a DNA molecule is taken up from the external environment and incorporated into the genome of the recipient cell or uptake of naked DNA by microbes.

132. L form bacteria

L-form organisms, are naturally found without the cell wall. In other words, normally, they have cell wall but under certain conditions, they may lack a cell wall.

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133. Function of cell wall in bacteria(3)

1) Shape 2) Protection from osmotic lysis 3) Pathogenicity

134. Define virus virusoid viroids .

- **Viruses :** They are acellular structure structure that consist of a protein capsid enclosing either DNA and sometime may contain an extra membrane called envelope .

a. **VIROIDS**: They are composed only of circular single stranded (ss) RNAV

b. **VIRUSOIDS**: They contain circular ssRNA that need helper viruses for replication and encapsidation. They are also called satellite viruses

c. **PRIONS**: These are basically infectious proteins formed inside the body. They behave like organisms as they can be transmitted from one individual to another. Extremely resistant to heat.

135. DNA lateral transfer

Lateral gene transfer or Horizontal gene transfer: - When genes are transferred from cells to cells within the same WV species, the process is called horizontal gene transfer. This can happen between cells of the same species, or across different species of organisms.

136. Types of Spiral Bacteria?

Vibrio: curved rods. Spin: Hel igid Spirochete: Helical but flexible

137. Characteristic of bacteria

Forms of bacterial colonies like Punctiform, circular, filamentous etc. 1

Elevation of bacterial colonies like Flat, raised, convex etc.

Margins of Bacterial colonies like entire look circular, undulate, curled etc.

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138. Magnetosomes characteristics.

- Inclusions of iron oxide
- Surrounded by invaginations of plasma membrane "
- Present in G negative bacteria

139. what are the physical and chemical methods for making Smear on the slide

a. Physical method: Heat

b. Chemical Methods: Ethanol or formaldehyde

140. How fungi can be identified by biochemical testing, 5 marks

Fungal biochemical tests are tests performed to identify fungi.

The following are some systems currently being used for the identification of yeasts:

Carbohydrate fermentation studies: This tests whether a certain yeast ferments different carbohydrates.

- **Carbohydrate assimilation** studies: This tests whether yeast can aerobically utilize different carbon and nitrogen compounds with oxygen as their final electron acceptor.
- API 20C System: Freeze-dried sugars are placed into wells on a plastic strip, yeast basal medium suspended yeast is pipetted into wells, and incubated. Turbidity determines growth.

Uni-Yeast Tek System: Identifies yeasts based on their ability to ferment certain sugars

141. Explain types of algae 5 marks

Phaeophyta (Kelp)

1) Brownish in color 2) Cellulose and alginic acid cell walls 3) Multicellular 4) Chlorophyll *a* and *c*, xanthophylls 5) Store carbohydrates 6) Harvested for algin

Rhodophyta

1) Reddish in color 2) Cellulose cell walls 3) Most are multicellular 4) Chlorophyll *a* and *d*, phycobiliproteins 5) Store glucose polymer 6) Harvested for agar and carrageenan

Chlorophyta

1) Green algae 2) Cellulose cell walls 3) Unicellular or multicellular 4) Chlorophyll *a* and 5) Store glucose polymer

Bacillariophyta

Brownish in color 2) Pectin and silica cell walls 3) Unicellular 4) Chlorophyll *a* and *c*, carotene, xanthophylls **5. Store oil 6) Fossilized diatoms formed oil**

Dinoflagellata

1) Brownish in color 2) Cellulose in plasma membrane 3) Unicellular 4) Chlorophyll *a* and *c*, carotene, xanthins 5) Store starch 6) Some are symbionts in marine animals Neurotoxins cause paralytic shellfish poisoning

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142. Explain types of conidiophores 5 marks

Conidiospores

- **'Conidia**: in this form Chain of spores are formed at the end of a conidiphore for example Aspergillus
- **Arthroconidia**: in this form spores are formed by fragmentation of septate hypha for example Coccidioides immitus
- **Blastoconidia**: spores are formed in the result of budding of the parent cells for example Candida albicans
- **Chlamydioconidia**: spores formed by rounding and enlargement within a hyphal segment Example Candida albicans

143. Explain ELISA 5 marks

Enzyme Linked Immunosorbent Assay —ELISA

- Some antigen antibody reactions are not visible to the eyes.
- Such reactions are visualized indirectly using various probes.
- robes include enzymes, radioisotopes etc. Serological Testing
- Enzymes or radioisotopes are conjugated to antibodies
- Presence of enzymes detected by incubating with substrate that forms colored compounds

144. Microbial identification method. Give detail of two. 5 marks

1. **Enzyme Linked Immunosorbent Assay —ELISA**

- Some antigen antibody reactions are not visible to the eyes.
- uch reactions are visualized indirectly using various probes.
- Probes include enzymes, radioisotopes etc. Serological Testing
- Enzymes or radioisotopes are conjugated to antibodies

2. **Western Blotting**

Western Blotting: We can use this technique for the detection of antigen in the serum. Microbial proteins can be separated on SDS-PAGE by electrophoresis and the presence of these proteins can be detected by enzyme-tagged antibodies specific to those proteins. A color band will be seen where the specific protein (antigen) is present on the gel. Please remember that proteins in the gel are first transferred to a paper strip before they could be detected by specific antibodies.

145. Epidemiology and its methods 5

- The study of where and when diseases occur and how they are transmitted in a population

Descriptive Epidemiology: (recording the data about disease). It includes location and time of cases of disease gender , health, age of patients etc. (Basically, frequency and distribution of risk factors in population are r orded). Preva nce and incidence of disease are typical examples. Such studies can be retrospective . So, in summary, in descriptive epidemiology, information about an

Already existing disease is recorded.

Analytical Epidemiology: It relates to determining the cause of the disease. Analysis of data mode of transmission of diseases and means to prevent diseases also come in this discipline. Such studies can be done with case.

Control methods in which factors that have preceded the disease are determine d . A group of people who have the disease is compared with another group of people in the same location who are the free of disease

A Cohort method tracks two groups forward from exposure to outcome cohort study compares the experience of a group exposed to the factor with another group not exposed to the factor.

146. Define immunity, difference between innate and adaptive immunity in Detail (10 marks)

Immunity: Ability of the body to protect itself from harmful effects of diseases.

Susceptibility: lack of immunity which leads to easy establish ment of disease.

Types of Immunity

1) Innate (Natural):

- Present at birth: One is born with these defenses, so it is present without any exposure to the microbes. That" is the reason, it is non-specific.
- Rapid to respond: Because it is already there, it interacts with the microbes as soon as they enter the body .
- **Nonspecific:** It means that it does not differentiate if it is an E. coli or a Staph or salmonella body defene system against these entire pathogens act in a similar fashion. In other words, the same microfage can phagocytize E. coli, Staph or Salmonella at the same time (if these pathogens are present there).

• Does not increase in response with repeated exposures to the same agent. Therefore, it has no memory.
• First line of defense: Physical barriers (skin, mucus membrane) provide the first line of defense as they prevent the entry of the organisms by providing a physical barrier which if broken will let the microbes go inside.
• Second line of defense: Phagocytic cells and their secretions provide a second line of barriers for microbes. So, if microbes succeed in breaching or breaking the first barrier, macrophages, neutrophils and such other cells phagocytize these microbes and kill them before these can establish infection in the body.

2) Adaptive (Acquired): It is the third line of defense. Acquired by experience, specific to an organism.
• Specific: It means that antibodies made against E.coli will destroy E. coli only, and will not do any harm to a Staph for example.
• Have memory: It means if antibodies have once been made against E. coli, this exposure will be remembered by the body, and if a second encounter is made with the same type of E.coli again, the body would mounta heightened immune response against E. coli because it has memory cells developed in the body which upon second stimulation proliferate quickly and mount a greater defensive response. O
• Because of memory, body's immune response increase with every repeated exposure. It has a limit also!
• Slower to respond: For the first time exposure, the body's response is slower and takes a long time compared with the innate immunity which is immediate.
• **Adaptive Immune Respons**e can further be divided into:
o Humoral immune response (antibody response by B cells or lymphocytes)
Cellular immune response (T cells are made which kills microbial harboring cells).

147. Write About the five layers of endospore? (5) marks

- ExOsporium: A thin delicate outermost covering of the spore
- Spore coat: 2nd layer underneath the exosporium. It is thick and composed of several protein layers

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- Cortex: t is the 3rd layer from outside in. t has peptidoglycan in it
- Spore cell wall or core wall: Surrounds the protoplast or spore core
- Spore core: Contains nucleoid and ribosomes

148. Function of cell membrane in bacteria? (5) marks

- ❖ Selective permeable barrier
- ❖ Passive and Active Transport
- ❖ Respiration in microbes (bacteria)
- ❖ Photosynthesis in microbes
- ❖ Lipid synthesis
- ❖ Cell wal parts are transported by a molecule called bactoprenol whieh is present in the plasma membrane.
- ❖ PM has many receptors in it with which various ligands can bind and in tiate signals for gene expression etc

149. Write names of kingdom as per five kingdom system of classification. (5) marks

In 1969, five kingdom classification was proposed by Robert whittaker as under.

1. **Kingdom Monera**: The Kingdom Monera consists of prokaryotic, unicellular organisms. No nuclear membrane or membrane-bound organelles such as chloroplasts. Examples of Monera include Archaea and Bacteria.

BACTERIA: These are prokaryotes, usually single-celled organisms. Their cell wall is made of peptidoglycan.

These are true bacteria

ARCHEA: These are also prokaryotes but their cell wall lacks peptidoglycan (may have special lipids

2. **PROTISTS**: This group includes algae, protoz0a, slime molds and water molds.

- ✓ Algae: These are eukaryotes. Their cell wall consists of cellulose. These are photosynthetic and produce oxygen
- ✓ Protoz0a: These are also eukaryotes, unicellular and have no cell wall. They move by pseudopods, flaßella or cilia. Some are photosynthetic
- ✓ Slime Molds: They are fungus like organisms at one time that may assume a protozoon like nature at another time. Basically, they are single celled but aggregate to form multicellular reproductive structures.
- ✓ Water Molds: These are protists that grow on the surface of fresh water and humid soils. Also known as mildews and white rusts. They look like fungus.

3. FUNGI: Usually unicellular (but also multicellular), their cell wall consists of chitin

4. Plantae: Multicellular eukaryotic and Autotrophs, photosynthesis. Like plants

5. Animalia: Multicellular eukaryotic and heterotrophs

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149. Characteristics of Metachromatic granules (5) marks

1. Also called volutin, they stain red with certain dyes such as methylene blue. That is why they are called metachromatic (stain in different color as methylene blue gives blue color but the color on these granules is red)

2. Large inclusions

3. These inclusions contain inorganic phosphates. Inorganic phosphates are used up in ATP synthesis.

4. Characteristics of Corynebacterium diphtheria: This bacterium can be identified by the presence of these granules in it.

150. Pili structure and its functions (5) marks

Structure

1. These are hair-like structure composed of pilin, usually one to ten in number.

2. Longer than fimbriae

Function

3. Used for attachment to: • Host cells • Bacteria

4. Used for DNA transfer from on bacterium to another: Conjugation (Sex pili)

5. Also function in twitching Motility

6. Gliding Motility is also the function of the pili

151. Name of types of microscopy (5) marks

Types of Microscopes:

1.Darkfield microscopy

- **Uses a darkfield condenser having an opaque disk**
- **Used for unstained specimens suspended in liquids**

2. Phase contrast microscopy

- **Light rays can be in phase or**
- **out of phase**

3. **Fluorescent microscopy**

- **Uses fluorochoromes and UV light as illumination source**
- **Fluorochoromes: substances that absorb short wavelength of light and emit longer wavelength (visible).**

4. **Transmission electron microscopy**

- **Electron microscope (EM) makes use of electron beam as an alternative to light rays**
- **It is used to analyses the internal structure of the organisms like cell organelles etc**

5. Scanning electron microscopy

• It is modified type of Electron microscopy in which electron beam doesn't pass through the specimen and the entire structure can be seen under this microscope

152. Types of bacteria based on oxygen requirement for growth (5) marks

Based on the use and concentration of Oxygen, microbes can be classified into five groups:

1. **Obligate Aerobes:** Oxygen must be present for their growth. These organisms do not grow if O2 is not present in the medium. If you test these organisms in a tube containing nutrient agar, the growth will remain confined to the surface only. Mycobacterium tuberculosis is an example of such an organism.

2. **Facultative Anaerobes**: These organisms prefer aerobic respiration; however, if oxygen is not available, they can use anaerobic respiration or fermentative modes for generating ATPs.

3. **Obligate Anaerobes:** Although these organisms do not use oxygen, they cannot grow in the presence of oxygen. The reason is that oxygen creates toxic compounds which can kill these cells.

4. **Aerotolerant Aerobes**: These organisms do not use oxygen and also are not bothered by the presence of oxygen. In other words, they have a better system to dispose of toxic oxygen compounds. They will grow evenly throughout the tube of medium as seen in the accompanying diagram.

5. Microaerophiles:-Aerobes but 02 is required in low amounts. - Produce 02 toxic compounds if exposed tomore 02 Examples Listeria, Vibrio

153. Classifications of bacteria on the basis of optimal temperature? (5) marks

Organisms are basically classified into three groups based on the temp requirements:

1. **Psychrophiles**: These are further divided into strict psychrophiles and psychrotrophs:

• Psychrotrophs: Cold loving: 15 oC

• Psychrotrophs: Optimum temp is 20-30 OC. Food spoilage bacteria that can spoil food during refrigeration.

2. **Mesophiles**: 25 - 40 OC: These are the ones that cause diseases in animals and humans.

- Moderate temp loving organisms
- Optimum: 370C as this is the body temperature of humans and animals.

3. **Thermophiles**: 50 - 60 OC

- Heat loving
- Important in organic compost piles
- These are further divided into hyper thermophiles that grow optimally at 80 oC. They live in hot springs.

154. Name 5 types of microbiology. (5) marks

1) Bacteriology 2) Mycology 3) Medical Biology 4) Immunology 5) Food microbiology

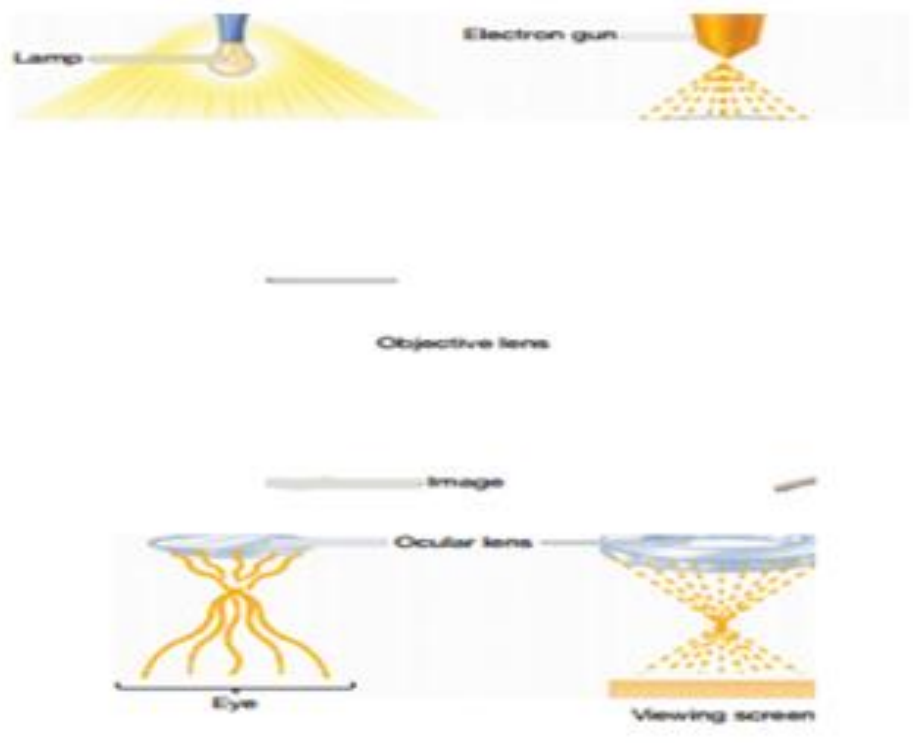
156. . Classes of media based (5) marks

Defined and complex media

- General purpose media
- Enriched media
- Selective media
- Differential media

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157. Label Electron Microscope (5) marks



QUESTIONS OF 2 AND 3 MARKS

1. Write economical uses of fungi.
2. What is sporulation?
3. Differentiate between viable and direct count.
4. What is flagellum? Write its functions.
5. What is meant by gene transfer? Name the types.
6. Describe classification system proposed by Robert H Whittaker.
7. What is taxonomic hierarchy?
8. What is western blotting?
9. Enlist types of passive transport.
10. Name types of spiral bacteria.
11. Define passive movement.
12. What is function of lysozymes?
13. Write methods for viable count of bacterial colonies.
14. Characteristics of magnetosomes.

15. .what are flurochromes...
16. .importance of fungi..
17. .microbial control
18. .advantages of staining..
19. .metachromatic granules..
20. .transmissn of infection..
21. Micribial identifcattion method.
22. Transmission of microbes
23. passive movemnt
24. flagellm and function
25. Magnetostome
26. immunity and its types in detail
27. Function of lipid A in lipopolysaccrides:
28. .Glycoclyx, its composition and function
29. Characters tics of aerophiles
30. 1. L bacteria3marks
31. Write composition of alchic acid equation?

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QUESTIONS OF 5 AND 10 MARKS

1. Explain characteristics of aerophiles.
2. Explain methods of microbial control in detail.
3. Explain methods of microbial identifications.
4. How fungi can be identified by biochemical testing?
5. Label the diagram of Electron microscope with numbers mentioned.
6. Write functions of cell membrane of bacteria.
7. Write names of 5 types of bacteria with respect to temperature.
8. Write characteristics of metachromatic granules.
9. Explain five kingdom classification systems.
10. . Explain layers of endospore.
11. Enlist types of bacteria on the basis of oxygen.
12. Name five types of microscopy.
13. Explain ELISA.
14. Explain types of algae
15. Magnetostome
16. how disease can be transmitd
17. Any two microbial identification method
18. how disease can be transmitd
19. Any two microbial identification method



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